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Gagan Pratap Sir

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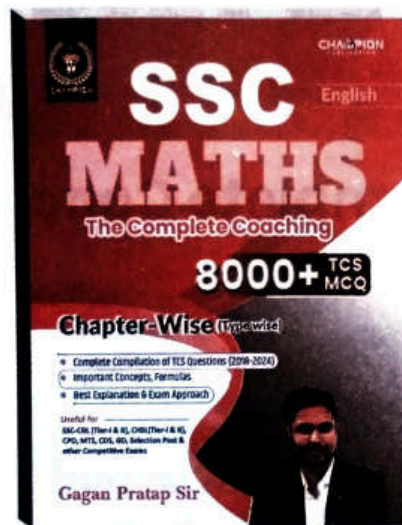
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Dedicated To God

My source of inspiration
and strength, I dedicate this book with heartfelt
gratitude. His unwavering guidance has
transformed my passion for writing.
His presence ignited my courage and clarity.
May this work resonate with those seeking
guidance and knowledge just as his light
has illuminated my path.
Thank you for the gift of
creativity and wisdom



Preface

The main aim of publishing this book is to spread the knowledge in the easiest way amongst learners. There are manifold purposes of writing this book on the subject. Basically, it caters to the needs of the candidates aspiring for competitive examinations, and for the beginners to understand the intricacies of the subject.

It is observed that the very name of the subject, Mathematics evokes fear in the minds of the students. Through this book an effort has been made to dispel that fear. MCQs at the end of every unit will help the students to make a self-assessment of the knowledge assimilated by going through the Chapters. The answers have been given for MCQs along with explanations.

As a professional it is observed that the books in the market are providing useful information to the students. They provide very useful multiple choice questions with their correct answers. The inquisitive mind of the student is still left high and dry as he is at a loss to know as to why a particular answer and why not otherwise. Through this book, the author has made an effort to provide rationale for the solutions. The book, therefore, meets the expectations of the students as it answers the demand and the quest in their mind.

The book is user-friendly and provides content in a well structured manner. It provides comprehensive and critical study of the various concepts of the subject matter. A word or suggestion from your side may add another feather to the cap of the subject matter of the book. The author looks forward to the comments, suggestions and criticism from the readers. Constructive suggestions and feedback from users would be highly appreciated, gratefully acknowledged and suitably incorporated.

Striving to serve the student community and to impart quality education.

With best wishes
Champion Publication



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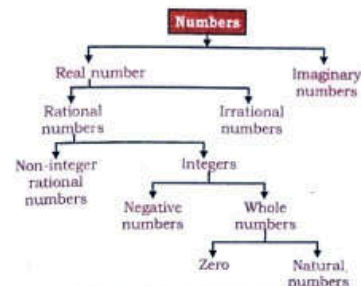
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NUMBER SYSTEM

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Classification of Numbers

Real Numbers which can be denoted on number line.

Imaginary Numbers can not be denoted on number line.

Ex. $+3, -7, 5, \frac{19}{13}, \frac{5}{7}$ $\sqrt{-7}, \sqrt{-3}, \sqrt{-5}, \sqrt{-1} = i$
 $0.0675, \sqrt{5}, \sqrt{11}$ $\downarrow$ $-1 = i^2$
 $a+ib$ $5+3i$

Real Numbers

Rational Numbers which can be written in

Irrational Numbers can not be written in

$\frac{p}{q}$ form ($q \neq 0$) $p, q \rightarrow \text{Int}$ $\frac{p}{q}$ form.

Ex. $\frac{5}{3}, \frac{13}{1}, \frac{-8}{1}, 0.5555, 0.1342607532 \dots$
 $\frac{22}{7}, \sqrt{q}, \pi = 3.141592 \dots$

Decimal Numbers

Terminating decimal	Non-Terminating Repeating decimal	Non-terminating Non-repeating decimal
$0.5 = \frac{1}{2}$	$0.333333\dots = \frac{1}{3}$	$\sqrt{2} = 1.414\dots$
$0.73 = \frac{73}{100}$	$0.565656\dots = \frac{56}{99}$	$\downarrow$
$0.648 = \frac{81}{125}$	$0.137137137\dots = \frac{137}{999}$	Irrational Numbers
Rational Numbers		

◊ Integers $\rightarrow$ All integers are rational number.

$\left(\frac{p}{q}\right)$ where $q = 1$

Integers

Negative Integers

$\{-\infty, \dots, -4, -3, -2, -1\}$
 $0 \rightarrow$ Neither positive nor negative

Non-negative Integers

$\{0, 1, 2, 3, 4, \dots, \infty\}$
 $\downarrow$
 Whole Numbers

◊ Natural numbers $\rightarrow \{1, 2, 3, 4, 5, \dots, \infty\}$

Integers

Even

Odd

Even $\rightarrow$ which are divisible by 2. (2K form)
 $\{0, 2, 4, 6, 8\}$

Odd $\rightarrow$ which are not divisible by 2. (2K + 1 form)
 $\{1, 3, 5, 7, 11\}$

◊ Odd $\times$ odd $\rightarrow$ odd
 odd $\times$ even $\rightarrow$ even
 odd $\div$ odd $\rightarrow$ even
 even $\div$ even $\rightarrow$ even
 odd $\div$ even $\rightarrow$ odd

$a + b = \text{odd} \rightarrow a = \frac{\text{odd} + \text{odd}}{2} = \frac{\text{even}}{2} = \text{Natural No.}$
 $a - b = \text{odd} \rightarrow a = \frac{\text{even} + \text{even}}{2} = \frac{\text{even}}{2} = \text{Natural No.}$

$a + b = \text{even} \rightarrow a = \frac{\text{even} + \text{even}}{2} = \frac{\text{even}}{2} = \text{Natural No.}$
 $a - b = \text{even} \rightarrow a = \frac{\text{even} + \text{odd}}{2} = \frac{\text{odd}}{2} \neq \text{natural No.}$

$a + b = \text{even} \rightarrow a = \frac{\text{even} + \text{odd}}{2} = \frac{\text{odd}}{2} \neq \text{natural No.}$
 $a - b = \text{odd} \rightarrow a = \frac{\text{even} + \text{odd}}{2} = \frac{\text{odd}}{2} \neq \text{natural No.}$

Natural numbers

Prime numbers

Composite numbers

Prime Numbers $\rightarrow$ Only two factors 1 & itself.
 2, 3, 5, 7, 11, 13, 23, 61, 67, 97 etc.
 2 $\rightarrow$ even prime no. & smallest prime no.
 3, 5, 7 $\rightarrow$ only pair of consecutive odd prime no.

Prime no. $\rightarrow$ 1-50 $\rightarrow$ 15
 50-100 $\rightarrow$ 10
 1-100 $\rightarrow$ 25
 1-200 $\rightarrow$ 46
 1-1000 $\rightarrow$ 168

Each prime number can be written in $(6k \pm 1)$ form. But every $(6k \pm 1)$ form may not be necessarily prime no.

$13 \rightarrow 6 \times 2 + 1$ (prime)
 $25 \rightarrow 6 \times 4 + 1$ (not prime)

❖ **Composite Numbers** $\rightarrow$ more than two factors.

Ex. 4, 6, 8, 9 etc.

1 $\rightarrow$ Neither prime nor composite

4 $\rightarrow$ Smallest composite number.

9 $\rightarrow$ Smallest odd composite number.

❖ **Relatively prime/co-prime numbers** $\rightarrow$ Two numbers in which nothing is common i.e. their HCF = 1
 (25, 19) (16, 9) (2, 3) (11, 13)

❖ **Twin-prime numbers** $\rightarrow$ Two prime numbers with a gap of 2.

(3, 5) (5, 7) (11, 13)

❖ only pair of prime no. with a gap of 2 is 3, 5, 7.

❖ Smallest 3 digit prime $\rightarrow$ 101

largest 3 digit prime $\rightarrow$ 997

Perfect Numbers

❖ If the sum of all the factors (excluding that no.) is equal to that number then it is called a perfect no.

$6 \rightarrow 1, 2, 3, 6$ (factors) $\therefore 1+2+3 = 6$

(Smallest perfect no.) $\therefore 6$ is perfect no.

$28 \rightarrow 1, 2, 4, 7, 14, 28$

$(1+2+4+7+14) = 28$

Perfect numbers $\rightarrow 6, 28, 496, 8128 \dots$

Divisibility Rules

❖ 1 is not divisible by any number except 1 but 1 is a universal factor.

Divisibility Rule of 2, 4, 8, 16

2 $\rightarrow$ Last digit should be divisible by 2.

4 $\rightarrow$ Last 2 digit should be divisible by 4.

8 $\rightarrow$ Last 3 digit should be divisible by 8.

16 $\rightarrow$ Last 4 digits should be divisible by 16

Divisibility Rule of 3 and 9

3 $\rightarrow$ Sum of digits should be divisible by 3.

9 $\rightarrow$ Sum of digits should be divisible by 9.

Divisibility Rule of 5, 25, 125

5 $\rightarrow$ Last digit should be 0 or 5

25 $\rightarrow$ Last two digit should be divisible by 25.

125 $\rightarrow$ Last 3 digit should be divisible by 125.

Divisibility Rule of 6

6 $\rightarrow 6 = 2 \times 3$ (co-prime factors)

$\therefore$ If a number is divisible by 2 & 3 both, that number will also be divisible by 6

Divisibility Rule of 7, 11, 13

5922 $\rightarrow$ Make pair of 3 digits from RHS

$\rightarrow$ Add alternate pairs & take difference

$\rightarrow$ If difference is divisible by 7, 11, 13 then number will be divisible by 7, 11, 13 respectively

6489 $\rightarrow 489 - 6 \rightarrow 483$

380247 $\rightarrow 380 - 247 \rightarrow 133$

❖ ABAB $\rightarrow$ divisible by 101

$73 \times 101 = 7373$

❖ ABCABC $\rightarrow$ divisible by 1001

$687 \times 1001 = 687687$

❖ $7 \times 11 \times 13 = 1001$

(Remember)

Divisibility Rule of 11

❖ If the difference between the sum of the digits at odd places and sum of the digits at even places is zero or multiple of 11.

Add even place digits $\rightarrow$ Take diff. $\rightarrow$ If diff. is 0 or multiple of 11 then no. will be divisible by 11
 Add odd place digits

$166452 \rightarrow 1+6+5 = 12$

$6+4+2 = 12$ diff. = 0 $\therefore$ div. by 11

$7945938 \rightarrow 28 - 17 = 11$

Divisibility Rule of 12

$12 = 4 \times 3$

❖ If a number is divisible by 4 and 3 both then that number will also be divisible by 12.

$72 \rightarrow 8 \times 9$

$88 \rightarrow 8 \times 11$

Remainder Theorem

Dividend / भाज्य
 Divisor $\leftarrow 13 \overline{) 72} \begin{matrix} 5 \\ 65 \\ 7 \end{matrix} \rightarrow$ quotient / भाजफल
 भाजक
 7 $\rightarrow$ Remainder / शेष

$72 = 13 \times 5 + 7$

Dividend = Divisor $\times$ Quotient + Remainder

$\frac{206}{11}, R = 8$

Remainder is always less than divisor.

❖ a, b, n $\rightarrow$ natural numbers

$a^n + b^n$ (n odd) $\rightarrow a^3 + b^3 = (a+b)(a^2 - ab + b^2)$

$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \rightarrow (a+b)(a^2 - ab + b^2)$

$(+ - + - + -)$ (If it starts with + then it is followed by -)

$(a^5 + b^5) = (a+b)(a^4 - a^3b + a^2b^2 - ab^3 + b^4)$

$= (a+b)(a^4 - a^3b + a^2b^2 - ab^3 + b^4)$

$\therefore a^n + b^n \rightarrow n$ odd $\rightarrow (a+b)$ is a factor always.

No. of the form Div. by (a+b) Div. by (a-b)

1. $a^n + b^n$ (n $\rightarrow$ odd) $\checkmark$ $\times$

2. $a^n + b^n$ (n $\rightarrow$ even) $\times$ $\times$

3. $a^n - b^n$ (n $\rightarrow$ odd) $\times$ $\checkmark$

4. $a^n - b^n$ (n $\rightarrow$ even) $\checkmark$ $\checkmark$

❖ If Power is odd $\rightarrow$

$a^n + b^n + c^n + d^n$ is divisible by (a+b+c+d)

Fermat's Theorem

❖ **Fermat's Theorem** $\rightarrow \frac{a^{p-1}}{p} = 1$ (Remainder)

p = prime number

a, p $\rightarrow$ co-prime

$\frac{50^{16}}{17} \rightarrow R = 1$

❖ $\frac{(ax+k)^n}{a} = \text{Remainder} \rightarrow K^n$

$\frac{(ax+1)^n}{a} \rightarrow \text{Remainder} = 1^n = 1$

$\frac{(ax-k)^n}{a} \rightarrow \text{Remainder} = (-k)^n$

$\frac{(ax-1)^n}{a} \rightarrow \text{Remainder} = (-1)^n$

n = even $\rightarrow$ 1
 n = odd $\rightarrow$ -1

Consecutive Remainder

❖ $17 \overline{) 8697} \begin{matrix} 511 \\ \underline{85} \\ 19 \\ \underline{17} \\ 27 \\ \underline{17} \\ 10 \end{matrix}$

consecutive remainder = 1, 2, 10

Divisor $\rightarrow$ HCF [85, 17, 17]

$\rightarrow 17$

8. **Successive Division**

Divide 620 by 8, 5, 6 successively

$8 \overline{) 620} \begin{matrix} 77 \\ \underline{56} \\ 60 \\ \underline{56} \\ 4 \end{matrix} \quad 5 \overline{) 77} \begin{matrix} 15 \\ \underline{5} \\ 27 \\ \underline{25} \\ 2 \end{matrix} \quad 6 \overline{) 15} \begin{matrix} 2 \\ \underline{12} \\ 3 \end{matrix}$

successive remainders $\rightarrow 4, 2, 3$

OR $\frac{8 \overline{) 620} \begin{matrix} 77 \\ \underline{56} \\ 60 \\ \underline{56} \\ 4 \end{matrix}}{5 \overline{) 77} \begin{matrix} 15 \\ \underline{5} \\ 27 \\ \underline{25} \\ 2 \end{matrix}} \downarrow$ Successive remainders
 $\frac{6 \overline{) 15} \begin{matrix} 2 \\ \underline{12} \\ 3 \end{matrix}}{3 \overline{) 2} \begin{matrix} 0 \\ \underline{3} \end{matrix}} \downarrow$ Final quotient

Factors

❖ Factors are the positive integers that can divide a number exactly.

Properties of factors

- 1 is a factor of every number.
- Every natural number is a factor of itself.
- Apart from 1 all natural numbers have atleast two factors.

Number of Factors (NOF)

❖ $12 \rightarrow 12, 24, 36, 48, 60 \dots$ (multiples)
 $12 \rightarrow 1, 2, 3, 4, 6, 12$ (factors)

NOF of 12 $\rightarrow 6$

perfect square NOF $\rightarrow 2$

Even NOF of 12 $\rightarrow 4$

perfect cube NOF $\rightarrow 1$

odd NOF of 12 $\rightarrow 2$

NOF of multiple of 3 $\rightarrow 3$

❖ $72 \rightarrow 2^3 \times 3^2$ (write in prime base)

$[2^3 2^1 2^1 2^1] [3^2 3^1 3^1]$

OR $4 \times 3 = 12$ combinations

$\therefore$ NOF $\rightarrow 12$

Direct: NOF = (power + 1) $\times$ (power + 1) $\dots$

NOF of 72 $\rightarrow (3+1) \times (2+1) \rightarrow 4 \times 3 \rightarrow 12$

❖ $N = a^x \times b^y \times c^z$

a, b, c $\rightarrow$ prime number

NOF of N = (x+1)(y+1)(z+1)

Sum of factors (SOF)

❖ $72 \rightarrow [2^3 + 2^2 + 2^1 + 2^0] \times [3^2 + 3^1 + 3^0] = 15 \times 13 = 195$

❖ $2160 \rightarrow 2^4 \times 3^3 \times 5^1$

SOF $\rightarrow [2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0] \times [3^4 + 3^3 + 3^2 + 3^1 + 3^0] \times [5^2 + 5^1 + 5^0]$

$\rightarrow 31 \times 40 \times 6 = 7440$

OR **Sum of factors** $\rightarrow \frac{a^r - 1}{(r-1)} \rightarrow GP$

$\therefore \frac{1(2^5 - 1)}{2 - 1} \times \frac{1(3^4 - 1)}{3 - 1} \times \frac{1(5^2 - 1)}{5 - 1}$

$\rightarrow 31 \times 40 \times 6 = 7440$

❖ $2160 = 2^4 \times 3^3 \times 5^1$

Even factors $\rightarrow$ minimum 2^1

NOF = $5 \times 4 \times 2 = 40$

$(2^0 2^1 2^2 2^3 2^4) \times (3^0 3^1 3^2 3^3) \times (5^0 5^1)$

Even NOF $\rightarrow 4 \times 4 \times 2 \rightarrow 32$

OR $2160 = 2^4 \times 3^3 \times 5^1 \rightarrow 2(2^3 \times 3^3 \times 5^1)$

$\downarrow$
 $4 \times 4 \times 2 = 32$

Sum of even factors $\rightarrow 30 \times 40 \times 6 \rightarrow 7200$

No. of odd factors $\rightarrow 40 - 32 = 8$

No. of odd factors $\rightarrow$ Absence of 2

$\therefore 2160 = 2^4 \times 3^3 \times 5^1 \rightarrow 4 \times 2 = 8$

Sum of odd factors $\rightarrow 40 \times 6 = 240$

❖ $2160 = 2^4 \times 3^3 \times 5^1$

$(2^0 2^1 2^2 2^3 2^4) (3^0 3^1 3^2 3^3) (5^0 5^1)$

Number of factors which are multiple of 18 $\rightarrow$

$18 = 2^1 \times 3^2$

$\therefore$ minimum 2^1 and 3^2 required for multiple of 18.

$\rightarrow 4 \times 2 \times 2 = 16$

OR $2160 = 2^4 \times 3^3 \times 5^1 \rightarrow 2 \times 3^2 [2^3 \times 3^1 \times 5^1]$

$\downarrow$
 $4 \times 2 \times 2 = 16$

❖ Sum of reciprocal of all factors = $\frac{\text{sum of factors}}{\text{number}}$

$$8 \rightarrow 1, 2, 4, 8$$

$$\text{Sum of reciprocal} \rightarrow \frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{8+4+2+1}{8}$$

$$= \frac{\text{Sum of factors}}{\text{number}}$$

❖ $10800 \rightarrow 2^4 \times 3^3 \times 5^2$

Number of factors which are perfect squares $\rightarrow$

a^{2m} = perfect square number

a^{3n} = perfect cube number

a^{6n} = perfect square as well as perfect cube

$$(2^0 2^2 2^4) \times (3^0 3^2) \times (5^0 5^2)$$

$$\rightarrow 3 \times 2 \times 2 = 12$$

OR No. of factors (perfect squares) $\rightarrow$

$$\frac{\text{Power}}{2} \rightarrow \text{Integer} + 1$$

$$\rightarrow (2+1)(1+1)(1+1)$$

$$\rightarrow 3 \times 2 \times 2 \rightarrow 12$$

❖ $72 \rightarrow 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72$

make pairs from 1st & last, NOF = 12

$$1 \times 72 = 72 \quad 4 \times 18 = 72$$

$$2 \times 36 = 72 \quad 6 \times 12 = 72$$

$$3 \times 24 = 72 \quad 8 \times 9 = 72$$

$$\text{Product of factor} = 72^6$$

$$\therefore \text{Product of factors of } N = (N)^{\frac{\text{NOF}}{2}}$$

❖ Only perfect square number has odd number of factors.

❖ Perfect square number from 1 to 100 $\rightarrow 10$

$$1, 4, 9, 16, 25, 36, 49, 64, 81, 100$$

$$\therefore \text{Even number of factors} \rightarrow 100 - 10 = 90$$

❖ Perfect square of a prime number has exactly 3 factors.

$$\begin{array}{c} 49 \rightarrow 7^2 \rightarrow 2 + 1 = 3 \\ 1 \quad \swarrow \quad \searrow \quad 49 \\ \quad 7 \end{array}$$

$$(1-100) \rightarrow 2^2, 3^2, 5^2, 7^2 \rightarrow 4 \text{ numbers}$$

❖ Numbers which have exactly 2 factors upto 100 = 25 (Prime Numbers)

❖ Numbers which have exactly 3 factors upto 100 = 4 (4, 9, 25, 49)

❖ Numbers which have more than 3 factors between (1 to 100) = ?

$$\text{only 1 factor} = 1 \quad (1 \text{ has only one factor})$$

$$\therefore \text{More than 3 factors} \rightarrow$$

$$100 - (1 + 4 + 25) = 70 \text{ numbers}$$



Number System

(Previous Year Practice Questions)



Divisibility Rule

SSC CGL

1. If a nine-digit number $985x3678y$ is divisible by 72, then the value of $(4x - 3y)$ is :

(CGL 2018 Pre)

- (a) 5 (b) 4
(c) 3 (d) 6

2. If a 10-digit number $2094x843y2$ is divisible by 88, then the value of $(5x - 7y)$ for the largest possible value of x , is :

(CGL 2018 Pre)

- (a) 3 (b) 5
(c) 2 (d) 6

3. If the 10-digit number $897359y7x2$ is divisible by 72, then what is the value of $(3x - y)$, for the possible greatest value of y ?

(CGL 2018 Pre)

- (a) 3 (b) 8
(c) 7 (d) 5

4. If an 11-digit number $5y5884805x6$, $x \neq y$, is divisible by 72, then the value of $\sqrt{xy}$ is :

(CGL 2018 Pre)

- (a) $\sqrt{7}$ (b) 3
(c) 7 (d) $\sqrt{6}$

5. If the six digit number $15x1y2$ is divisible by 44, then $(x + y)$ is equal to :

(CGL 2018 Pre)

- (a) 8 (b) 7
(c) 6 (d) 9

6. What is the least value of x such that $517x324$ is divisible by 12?

(CGL 2018 Pre)

- (a) 3 (b) 1
(c) 0 (d) 2

7. When 732 is divided by a positive integer x , the remainder is 12. How many values of x are there?

(CGL 2019 Pre)

- (a) 19 (b) 20
(c) 18 (d) 16

8. If the 6-digit numbers $x35624$ and $1257y4$ are divisible by 11 and 72, respectively, then what is the value of $(5x - 2y)$?

(CGL 2019 Pre)

- (a) 14 (b) 12
(c) 10 (d) 13

9. If nine-digit number $708x6y8z9$ is divisible by 99, then what is the value of $x + y + z$?

(CGL 2019 Pre)

- (a) 27 (b) 5
(c) 16 (d) 9

10. The largest number which should replace * in the number $2365*4$ to make the number divisible by 4 is :

(CGL 2019 Pre)

- (a) 9 (b) 0
(c) 2 (d) 8

11. If a five digit number $247xy$ is divisible by 3, 7 and 11, then what is the value of $(2y - 8x)$?

(CGL 2020 Pre)

- (a) 6 (b) 17
(c) 9 (d) 11

12. Find the difference between squares of the greatest value and the smallest value of P if the number $5306P2$ is divisible by 3.

(CGL 2020 Pre)

- (a) 60 (b) 68
(c) 36 (d) 6

13. Find the sum of squares of the greatest value and the smallest value of K in the number so that the number $45082K$ is divisible by 3.

(CGL 2020 Pre)

- (a) 68 (b) 64
(c) 100 (d) 50

14. The number $823p2q$ is exactly divisible by 7, 11 and 13. What is the value of $(p - q)$?

(CGL 2020 Pre)

- (a) 8 (b) 3
(c) 5 (d) 11

15. If the 5-digit number $593ab$ is divisible by 3, 7 and 11, then what is the value of $(a^2 - b^2 + ab)$?

(CGL 2020 Pre)

- (a) 35 (b) 31
(c) 25 (d) 29

16. The sum of 3-digit numbers abc , cab and bca is not divisible by :

(CGL 2020 Pre)

- (a) $a + b + c$ (b) 37
(c) 31 (d) 3

17. Find the sum of all the possible values of $(a + b)$, so that the number $4a067b$ is divisible by 11.

(CGL 2020 Pre)

- (a) 5 (b) 16
(c) 21 (d) 11

18. Find the greatest number $23a68b$, which is divisible by 3 but NOT divisible by 9.

(CGL 2021 Pre)

- (a) 238689 (b) 239685
(c) 239688 (d) 237687

19. If the 7-digit number $x8942y4$ is divisible by 56, what is the value of $(x^2 + y)$ for the largest value of y , where x and y are natural numbers?

(CGL 2021 Pre)

- (a) 33 (b) 44
(c) 55 (d) 70

20. If the 9-digit number $7x79251y8$ is divisible by 36, What is the value of $(10x^2 - 3y^2)$ for the largest possible value of y ?

(CGL 2021 Pre)

- (a) 490 (b) 289
(c) 192 (d) 298

21. If the nine-digit number $9m2365n48$ is completely divisible by 88, what is the value of $(m^2 \times n^2)$, for the smallest value of n , where m and n are natural numbers?

(CGL 2021 Pre)

- (a) 36 (b) 64
(c) 32 (d) 20

22. Find the greatest number $234a5b$, which is divisible by 22, but NOT divisible by 5.

(CGL 2021 Pre)

- (a) 234058 (b) 234850
(c) 234652 (d) 234751

23. Find the value of k in the number $3426k$ if the number is divisible by 6 but NOT divisible by 5.

(CGL 2021 Pre)

- (a) 4 (b) 6
(c) 3 (d) 9

24. Which of the following pairs of non-zero values of p and q make 6-digit number $674pq0$ divisible by both 3 and 11?

(CGL 2022 Pre)

- (a) $p = 2$ and $q = 2$
(b) $p = 5$ and $q = 4$
(c) $p = 4$ and $q = 2$
(d) $p = 5$ and $q = 2$

25. $3^{50} + 9^{25} + 27^{18} + 9^{28} + 9^{29}$ is divisible by which of the following integers?

(CGL 2022 Pre)

- (a) 11 (b) 5
(c) 7 (d) 2

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26. If a number $K = 42 \times 25 \times 54 \times 135$ is divisible by 3a, then find the maximum value of a?
(CGL 2022 Pre)
(a) 6 (b) 7
(c) 4 (d) 5
27. If 7-digit number 678p37q is divisible by 75 and p is not a composite, then the values of p and q are:
(CGL 2022 Pre)
(a) p = 5, q = 5
(b) p = 3, q = 0
(c) p = 3, q = 5
(d) p = 2, q = 5
28. Any six-digit number that is formed by repeating a three-digit number, is always divisible by:
(CGL 2022 Pre)
(a) 111 (b) 1001
(c) 19 (d) 101
29. If the seven-digit number 52A6B7C is divisible by 33, and A, B, C are primes, then the maximum value of $2A+3B+C$ is:
(CGL 2022 Pre)
(a) 32 (b) 23
(c) 27 (d) 34
30. $6^{28} + 6^{26} + 6^{27} + 6^{28}$ is divisible by:
(CGL 2023 Pre)
(a) 256 (b) 254
(c) 255 (d) 259
31. Which of the following numbers is divisible by 36?
(CGL 2023 Pre)
(a) 47502 (b) 29412
(c) 54732 (d) 87064
32. Which of the numbers 9592450, 9592330, 9592885 and 9592741 is divisible by 11?
(CGL 2023 Pre)
(a) 9592885 (b) 9592741
(c) 9592450 (d) 9592330
33. Which number among 34936, 35508, 35580 and 36508 is divisible by 33?
(CGL 2023 Pre)
(a) 35508 (b) 35580
(c) 36508 (d) 34936
34. Which of the following numbers is divisible by 99?
(CGL 2023 Pre)
(a) 31548 (b) 60687
(c) 44775 (d) 84456
35. Which of the following numbers is divisible by 44?
(CGL 2023 Pre)
(a) 32802 (b) 54736
(c) 93472 (d) 27048
36. A four-digit number abba is divisible by 4 and $a < b$. How many such numbers are there?
(CGL 2023 Pre)
(a) 10 (b) 8
(c) 12 (d) 6
37. A 6-digit number has digits as consecutive natural numbers. The number is always divisible by _____.
(CGL 2023 Pre)
(a) 4 (b) 5
(c) 2 (d) 3
38. How many of the following numbers are divisible by 3 but NOT by 9?
(CGL 2023 Pre)
5826, 5964, 6039, 6336, 6489, 6564, 6867 and 6960
(a) 5 (b) 3
(c) 4 (d) 6
39. The five-digit number 45yz0 is divisible by 40. What is the maximum possible value of $(y+z)$?
(CGL 2024 Pre)
(a) 15 (b) 18
(c) 16 (d) 17
40. Replace * by the smallest digit so that $723*56*6$ is divisible by 6.
(CGL 2024 Pre)
(a) 4 (b) 3
(c) 2 (d) 1
41. If $321y72$ is a multiple of 6, where y is a digit, what is the least value of y?
(CGL 2024 Pre)
(a) 3 (b) 0
(c) 7 (d) 2
42. If 436P5 is divisible by 3, then the largest value of P is:
(CGL 2024 Pre)
(a) 12 (b) 6
(c) 8 (d) 9
43. The number 4,29,714 is divisible by:
(CGL 2024 Pre)
(a) 3 and 5 (b) 6 and 5
(c) 4 and 6 (d) 3 and 6
44. Which composite number can divide the sum of the first 12 natural numbers?
(CGL 2024 Pre)
(a) 6 (b) 12
(c) 4 (d) 8
45. $7^{15} + 7^{16} + 7^{17}$ is divisible by which of the following given numbers?
(CGL 2024 Pre)
(a) 3 (b) 5
(c) 4 (d) 4
46. For what least value of n, $2^{2n} + 2n$ is completely divided by 6, where n is an integer?
(CGL 2024 Pre)
(a) 3 (b) 1
(c) 0 (d) 2
47. If m is even, then $(8^m - 1)$ is divisible by:
(CGL 2024 Pre)
(a) 63 (b) 42
(c) 65 (d) 8
48. From the given numbers A, B, C and D, which number is NOT divisible by 11?
 $A = 712712$, $B = 177210$, $C = 64614$, $D = 756148$
(CGL 2024 Pre)
(a) C (b) D
(c) B (d) A
49. Find the greatest value of 'k' in a 6-digit number 6745k2 such that the number is divisible by 3.
(CGL 2024 Pre)
(a) 8 (b) 6
(c) 9 (d) 7
50. The least number which must be subtracted from 7278745 so as to obtain a sum that is divisible by 11 is:
(CGL 2024 Pre)
(a) 3 (b) 1
(c) 5 (d) 2
51. Which digits should come in the place of * and #, respectively, if the 7-digit number 62684*# is divisible by both 8 and 5?
(CGL 2024 Pre)
(a) 2 and 0 (b) 4 and 0
(c) 5 and 0 (d) 0 and 5
52. If 7845K854 is divisible by 11, then what is the value of K?
(CGL 2024 Pre)
(a) 6 (b) 9
(c) 7 (d) 8
53. Which of following numbers is NOT divisible by 11?
(CGL 2024 Pre)
(a) 109153 (b) 5445
(c) 128041 (d) 80124
54. Find the smallest value that must be assigned to number 'a' in order for 91876a2 to be divisible by 8.
(CGL 2024 Pre)
(a) 3 (b) 0
(c) 1 (d) 2
55. Which digits should come in place * and #, respectively, if the number 72864*\$ is divisible by both 8 and 5?
(CGL 2024 Pre)
(a) 4 and 0 (b) 2 and 0
(c) 2 and 5 (d) 4 and 5
56. Which of the following numbers is divisible by 2, 5, 10 and 11?
(CGL 2024 Pre)
(a) 203467 (b) 830942
(c) 589270 (d) 1234560
57. Find the least value of k for which a 7-digit number 249k876 is divisible by 6.
(CGL 2024 Pre)
(a) 1 (b) 0
(c) 5 (d) 7
58. Which of the following can be the value of 'k' so that the number 217924k is divisible by 6?
(CGL 2024 Pre)
(a) 4 (b) 6
(c) 2 (d) 0
59. Find the least value of x for which $57x716$ is divisible by 9.
(CGL 2024 Pre)
(a) 1 (b) 9
(c) 0 (d) 8
60. If P is a digit such that 6954P is divisible by 11, then P equals:
(CGL 2024 Pre)
(a) 2 (b) 8
(c) 6 (d) 7
61. Which of the following numbers is divisible by 6?
(CGL 2024 Pre)
(a) 128530 (b) 12581
(c) 438750 (d) 62233
62. Which of the following numbers is NOT divisible by 8?
(CGL 2024 Pre)
(a) 7344 (b) 5544
(c) 7497 (d) 4608
63. Which of the following numbers is divisible by both 2 and 5?
(CGL 2024 Pre)
(a) 63840 (b) 20792
(c) 37915 (d) 37254
64. If the seven digits number $56x34y4$ is divisible by 72, then what is the least value of $(x+y)$?
(CPO 2018 Pre)
(a) 8 (b) 12
(c) 5 (d) 14
65. The number 106974 is divisible by which of the single digit numbers:
(CPO 2018 Pre)
(a) 2, 3, 6 and 7 only
(b) 2, 3 and 7 only
(c) 2, 3 and 4 only
(d) 2 and 3 only
66. The number 45789 is divisible by which of the single digit numbers:
(CPO 2018 Pre)
(a) Only by 9
(b) Only by 3 and 9
(c) Only by 3
(d) Only by 3 and 7
67. If the six-digit number 479xyz is exactly divisible by 7, 11 and 13, then $((y+z) \div x)$ is equal to:
(CPO 2019 Pre)
(a) $\frac{11}{9}$ (b) 4
(c) $\frac{13}{7}$ (d) $\frac{7}{13}$
68. If the 5-digit number 538xy is divisible by 3, 7 and 11, then the value of $(x^2 + y^2)$ is:
(CPO 2019 Pre)
(a) 10 (b) 17
(c) 25 (d) 13
69. The number 1563241234351 is:
(CPO 2019 Pre)
(a) divisible by both 3 and 11
(b) divisible by 11 but not by 3
(c) divisible by 3 but not by 11
(d) neither divisible by 3 nor by 11
70. If six-digit number $5x2y6z$ is divisible by 7, 11 and 13, then the value of $(x-y+3z)$ is:
(CPO 2020 Pre)
(a) 9 (b) 7
(c) 4 (d) 0
71. If the 9-digit number $72x8431y4$ is divisible by 36, what is the value of $\left(\frac{x-y}{y-x}\right)$ for the smallest possible value of y, given that x and y are natural numbers?
(CPO 2022 Pre)
(a) $\frac{5}{7}$ (b) $\frac{2}{10}$
(c) $\frac{2}{5}$ (d) $\frac{9}{10}$
72. How many of the following numbers are divisible by 156? 312, 620, 936, 1402, 1872, 3216, 7176, 8108.
(CPO 2022 Pre)
(a) 5 (b) 3
(c) 4 (d) 1
73. If the number 6336633P is divisible by 132, then the value of P is:
(CPO 2022 Pre)
(a) 4 (b) 2
(c) 3 (d) 6
74. What are the values of R and M, respectively, if the given number is perfectly divisible by 16 and 11?
(CPO 2022 Pre)
 $34R05030M6$
(a) 4 and 6 (b) 7 and 5
(c) 5 and 5 (d) 5 and 7
75. If the following numbers are exactly divisible by 9, then what are the values of & # respectively?
(CPO 2022 Pre)
 $51\&918\#0$
(a) 4 and 5 (b) 8 and 4
(c) 9 and 8 (d) 2 and 2
76. What is the smallest perfect square number which is completely divisible by 4, 6, 9, 12 and 15?
(CPO 2023 Pre)
(a) 784 (b) 900
(c) 961 (d) 841
77. If the nine-digit number $43x1145y2$ is divisible by 88, then the value of $(3x-2y)$, for the smallest value of y, is:
(CHSL 2018 Pre)
(a) 22 (b) 18
(c) 20 (d) 9
78. If the eight-digit number $342x18y6$ is divisible by 72, then what is the value of $\sqrt{9x+y}$, for the largest value of y?
(CHSL 2018 Pre)
(a) $2\sqrt{7}$ (b) $4\sqrt{7}$
(c) 8 (d) 6
79. Which among the following numbers is exactly divisible by 11, 13 and 7?
(CHSL 2018 Pre)
(1) 624613 (2) 624624
(3) 624635 (4) 624646
(a) (1) (b) (2)
(c) (3) (d) (4)
80. The ten digit number $2x600000y8$ is exactly divisible by 24. If $x \neq 0$ and $y \neq 0$, then the least value of $(x+y)$ is equal to:
(CHSL 2018 Pre)
(a) 5 (b) 8
(c) 9 (d) 2
81. The 10-digit number $79x00001y6$ is exactly divisible by 88. What is the value of $(x+y)$?
(CHSL 2018 Pre)
(a) 5 (b) 9
(c) 6 (d) 7
82. If the number $59a44b$ is divisible by 36, then the maximum value of $a+b$ is:
(CHSL 2019 Pre)
(a) 16 (b) 12
(c) 14 (d) 10
83. Which are the two nearest numbers to 19,596 divisible by 9?
(CHSL 2019 Pre)
(a) 19,564; 19,620
(b) 19,611; 19,575
(c) 19,509; 19,611
(d) 19,593; 19,602

84. If m is a natural number, then $(7a^2 + 7a)$ is always divisible by:
(CHSL 2019 Pre)
(a) 7 & 14 both (b) 7 only
(c) 14 only (d) 21 only
85. Which of the following number is divisible by 2, 5 and 10?
(CHSL 2019 Pre)
(a) 7,20,345 (b) 1,25,372
(c) 19,400 (d) 149
86. Which of the following numbers is divisible by both 7 and 11?
(CHSL 2019 Pre)
(a) 16,324 (b) 12,235
(c) 16,257 (d) 16,425
87. If $4M37094267N$ is divisible by both 8 and 11, where M and N are single digit integers, then the values of M and N are:
(CHSL 2019 Pre)
(a) $M = 5, N = 6$
(b) $M = 2, N = 5$
(c) $M = 5, N = 2$
(d) $M = 5, N = 4$
88. If 8-digit number $4432A43B$ is divisible by 9 and 5, then the sum of A and B is equal to:
(CHSL 2019 Pre)
(a) 12 (b) 5
(c) 7 (d) 8
89. $2^{14} - 1$ is divisible by:
(CHSL 2019 Pre)
(a) 11 (b) 17
(c) 13 (d) 7
90. If the number $62783xy$ is divisible by both 8 and 5, then the smallest possible value of x and y is:
(CHSL 2019 Pre)
(a) $x = 2, y = 2$ (b) $x = 6, y = 0$
(c) $x = 2, y = 0$ (d) $x = 2, y = 5$
91. If the five-digit number $672xy$ is divisible by 3, 7 and 11, then what is the value of $(6x + 5y)$?
(CHSL 2020 Pre)
(a) 23 (b) 24
(c) 16 (d) 17
92. If a nine-digit number $489x6378y$ is divisible by 72, then the value of $\sqrt{8x+6y}$ will be:
(CHSL 2020 Pre)
(a) 10 (b) 4
(c) 6 (d) 8
93. If the number $645A2879B8$ is divisible by both 8 and 9, then the smallest possible values of A and B will be:
(CHSL 2020 Pre)
(a) $A = 3, B = 4$
(b) $A = 4, B = 3$
(c) $A = 2, B = 3$
(d) $A = 3, B = 2$

94. What is the sum of all the possible values of k for which a seven-digit number $23k567k$ is divisible by 3?
(CHSL 2020 Pre)
(a) 15 (b) 3
(c) 109 (d) 16
95. If the 11-digit number $4y6884805x6$ is divisible by 72, and $x \neq y$, then the value of $\sqrt{xy}$ is:
(CHSL 2020 Pre)
(a) $\sqrt{12}$ (b) $\sqrt{6}$
(c) $\sqrt{5}$ (d) $\sqrt{8}$
96. If the five-digit number $457ab$ is divisible by 3, 7 and 11, then what is the value of $a^2 + b^2 - ab$?
(CHSL 2020 Pre)
(a) 49 (b) 36
(c) 24 (d) 33
97. If the nine-digit number $8475639AB$ is divisible by 99, then what is the value of A and B ?
(CHSL 2020 Pre)
(a) $A=4, B=6$ (b) $A=5, B=3$
(c) $A=3, B=9$ (d) $A=4, B=8$
98. If the number $583p2310q2$ is divisible by 11, then what is the value of $p \times q$, where $p > q$?
(CHSL 2020 Pre)
(a) 4 (b) 2
(c) 0 (d) 6
99. Which of the following is divisible by 88?
(CHSL 2020 Pre)
(a) 2776408 (b) 2776400
(c) 2767440 (d) 2767416
100. If an eleven-digit number $6578x43267y$ is divisible by 72, then the value of $\sqrt{x+6y}$ will be:
(CHSL 2020 Pre)
(a) 3 (b) 5
(c) 6 (d) 4
101. The six-digit number $537xy5$ is divisible by 125. How many such six-digit numbers are there?
(CHSL 2020 Pre)
(a) 4 (b) 2
(c) 3 (d) 5
102. If the nine-digit number $48x4923y8$ is divisible by 88, then the value of $(6x + 5y)$ for the maximum value of y , will be:
(CHSL 2020 Pre)
(a) 76 (b) 72
(c) 65 (d) 71
103. From the following numbers, find the number which is exactly divisible by 42.
(CHSL 2021 Pre)
(a) 25232 (b) 25242
(c) 25244 (d) 25212

104. What will be the smallest natural number to be filled in the blank for the number '23_45678' to be divisible by 22?
(CHSL 2021 Pre)
(a) 7 (b) 1
(c) 9 (d) 3
105. If the 3-digit number $2x3$ is divisible by 3, then the value of $x^2 + x$ is (consider the maximum value of x):
(CHSL 2021 Pre)
(a) 7 (b) 20
(c) 56 (d) 2
106. If the number $732XY$ is divisible by 70, then find the minimum value of $\frac{x+y}{2}$.
(CHSL 2021 Pre)
(a) 2 (b) 1
(c) 0 (d) 3
107. Which of the following numbers is divisible by 15?
(CHSL 2021 Pre)
(a) 1,65,485 (b) 3,06,045
(c) 2,12,695 (d) 2,95,145
108. Which one of the following numbers is divisible by 24?
(CHSL 2021 Pre)
(a) 569896 (b) 478656
(c) 480128 (d) 543344
109. If the 9-digit number $5p42978n6$ is divisible by 72, what is the value of $(2p - 1)$, where n is the second largest of all the possible values of n ? Given that p and n are natural number.
(CHSL 2021 Pre)
(a) 17 (b) 21
(c) 15 (d) 11
110. Which of the following numbers is NOT divisible by 36?
(CHSL 2021 Pre)
(a) 136216 (b) 124344
(c) 259884 (d) 120816
111. Which of the following numbers is divisible by 55?
(CHSL 2021 Pre)
(a) 178765 (b) 185625
(c) 171125 (d) 164485
112. Which of the following numbers is NOT divisible by 150?
(CHSL 2021 Pre)
(a) 320550 (b) 333300
(c) 463750 (d) 201300
113. Which of the following numbers is NOT divisible by 75?
(CHSL 2021 Pre)
(a) 117975 (b) 275475
(c) 163750 (d) 666600

Number System

114. Which of the following numbers is divisible by 360?
(CHSL 2021 Pre)
(a) 171720 (b) 176600
(c) 166200 (d) 171080
115. Which of the following options is NOT divisible by 18?
(CHSL 2021 Pre)
(a) 571032 (b) 732546
(c) 245798 (d) 459018
116. Find the greatest possible value of $(A + B)$ for which the 8-digit number $143B203A$ is divisible by 15.
(CHSL 2021 Pre)
(a) 15 (b) 17
(c) 16 (d) 14
117. If $7A425B$ is divisible by 36, then what is the value of $A - B$?
(CHSL 2022 Pre)
(a) 0 (b) 5
(c) 1 (d) 2
118. If the 4-digit number $48ab$ is divisible by 2, 5 and 7, then what is the value of $(10a - b)$?
(CHSL 2022 Pre)
(a) 0 (b) 20
(c) 10 (d) 30
119. What is the value of x in the number $3426x$ if the number is divisible by 6 but NOT divisible by 5?
(CHSL 2023 Pre)
(a) 8 (b) 4
(c) 3 (d) 6
120. Find the lowest positive value of $(c - b)$ such that the 7-digit number $1738b9c$ is divisible by 12.
(CHSL 2024 Pre)
(a) 4 (b) 2
(c) 1 (d) 7
121. Write the smallest digit in the blank space of a number 7_7624 , so that the number formed is divisible by 3.
(CHSL 2024 Pre)
(a) 1 (b) 2
(c) 0 (d) 3

SSC MTS & SP

122. The 10-digit number $90457416xy$ is divisible by 9, and $x - y = 3$. What is the value of $(5x + 3y)$?
(SP Phase-8)
(a) 48 (b) 39
(c) 33 (d) 29
123. Find the number nearest to 51462, that is divisible by 8.
(SP Phase-9)
(a) 51464 (b) 51468
(c) 51456 (d) 51480

124. Find the sum of the greatest and the smallest number which may replace k in the number $8130k36$ so that the number is divisible by 8.
(SP Phase-9)
(a) 10 (b) 9
(c) 12 (d) 8
125. A six-digit number of the form $ababab$ is always divisible by:
(SP Phase-9)
(a) 11111 (b) 10001
(c) 11001 (d) 10101
126. If the five-digit number $570xy$ is divisible by 231, then what is the value of $(2x - y)$?
(SP Phase-9)
(a) 5 (b) 7
(c) 6 (d) 3
127. If a number N is divisible by 3, then which of the following is true?
(SP Phase-9)
(a) $(N + 6)$ is divisible by 6
(b) $(N + 2)$ is divisible by 3
(c) N^2 is divisible by 6
(d) $(N + 12)$ is divisible by 3
128. If the 9-digit number $957x5y6z3$ is divisible by 33, then what is the maximum value of $(x+y+z)$?
(SP Phase-9)
(a) 24 (b) 27
(c) 25 (d) 26
129. The sum of 3-digit numbers abc , bca and cab is always divisible by:
(SP Phase-9)
(a) 35 (b) 41
(c) 37 (d) 31
130. If a 9-digit number $609a5b9c9$ is divisible by 99, then $(a+b+c)$ is equal to:
(SP Phase-9)
(a) 16 (b) 7
(c) 26 (d) 25
131. A 4-digit number $x92x$ is divisible by 36 and a 7-digit number $y569x04$ is divisible by 11. For the maximum value of $(y + z)$, what is the value of $(2x + y + z)$?
(SP Phase-9)
(a) 31 (b) 22
(c) 33 (d) 26
132. If the 7-digit number $612x97y$ is divisible by 72, and the 6-digit number $34z178$ is divisible by 11, then the value of $(x-2y+3z)$ is:
(SP Phase-9)
(a) 7 (b) 5
(c) 2 (d) 6
133. Which of the following numbers will completely divide $(6^{61} + 6^{62} + 6^{63} + 6^{64})$?
(SP 2023 Phase-11 Graduation L)
(a) 7 (b) 13
(c) 10 (d) 11

134. The six-digit number $7x1yyx$ is a multiple of 33 for non-zero digits x and y . Which of the following could be a possible value of $(x + y)$?
(SP 2023 Phase-11 Matric. L)
(a) 4 (b) 5
(c) 2 (d) 3
135. If the product $6352 \times 7A1$ is divisible by 12, then the value of A is:
(SP 2022 Phase-10 Graduation L)
(a) 6 (b) 3
(c) 5 (d) 4
136. A five-digit number 58828 is divisible by 77. If we add 5 more than twice of 11 to the number, then the resultant number will be divisible by:
(SP 2022 Phase-10 Matric. L)
(a) 5 (b) 11
(c) 7 (d) 33
137. A seven-digit number 5702718 is divisible by 147. If we rearrange the digits of the number in descending order and subtract 4 more than three times of 17 from the new number which is formed, then the resultant number will be divisible by:
(SP 2022 Phase-10 Matric. L)
(a) 17 (b) 11
(c) 7 (d) 5
138. A number is square of 167. When digits of the number are rearranged in descending order, then the new number which is formed will be divisible by:
(SP 2022 Phase-10 Matric. L)
(a) 17 (b) 13
(c) 5 (d) 11
139. A number is cube of 53. When 7 times of 57 is subtracted from the number, then the resultant number which is formed will be divisible by:
(SP 2022 Phase-10 Matric. L)
(a) 12 (b) 19
(c) 13 (d) 11

SSC MAINS

140. $5^{71} + 5^{72} + 5^{73} + 5^{74} + 5^{75}$ is divisible by which of the following number?
(CGL MAINS 2019)
(a) 71 (b) 69
(c) 89 (d) 73

141. A number is divisible by 3 only when: (CHSL 2022 Mains)

- (a) the difference of the sum of the odd and the even digits is divisible by 3
- (b) the sum of its digits is divisible by 3
- (c) the last digit is either 0 or an even number
- (d) the last two digits are divisible by 3

142. $222^{333} + 333^{222}$ is divisible by which of the following numbers? (UPSC CSAT 2024)

- (a) 2 and 3 but not by 37
- (b) 3 and 37 but not 2
- (c) 2 and 37 but not 3
- (d) 2, 3 and 37

When divided by 2 or more numbers and leaves same remainder

SSC CGL

143. Find the greatest number which divides 108, 124 and 156, leaving the same remainder. (CGL 2021 Pre)

- (a) 18
- (b) 10
- (c) 12
- (d) 16

144. What is the greatest number by which when 156, 181 and 331 are divided, the remainder is 6 in each case? (CGL 2021 Pre)

- (a) 26
- (b) 17
- (c) 25
- (d) 13

145. What is the least number which when decreased by 7 is divisible by 15, 24, 28 and 32? (CGL 2021 Pre)

- (a) 10097
- (b) 10087
- (c) 10067
- (d) 10077

146. Find the greatest 3-digit number which, when divided by 3, 4, 5 and 8, leaves remainder 2 in each case. (CGL 2021 Pre)

- (a) 962
- (b) 122
- (c) 958
- (d) 482

147. The largest five-digit number which when divided by 7, 9 and 11, leaves the same remainder as 3 in each case, is: (CGL 2022 Pre)

- (a) 95840
- (b) 98685
- (c) 96720
- (d) 99795

SSC CPO

148. Let x be the least 4-digit number which when divided by 2, 3, 4, 5, 6 and 7 leaves a remainder of 1 in each case. If x lies between 2800 and 3000, what is the sum of the digits of x ? (CPO 2019 Pre)

- (a) 15
- (b) 16
- (c) 12
- (d) 13

149. Let x be the least number of 4 digits that when divided by (2, 3, 4, 5, 6 and 7 leaves a remainder of 1 in each case. If x lies between 2000 and 2500, then what is the sum of the digits of x ? (CPO 2019 Pre)

- (a) 9
- (b) 15
- (c) 10
- (d) 4

150. Let x be the least number divisible by 13, such that when x is divided by 4, 5, 6, 7, 8 and 12, the remainder in each case is 2. The sum of the digits of x is: (CPO 2019 Pre)

- (a) 11
- (b) 8
- (c) 9
- (d) 10

151. What is the greatest number of six digits, which when divided by each of 16, 24, 72 and 84, leaves the remainder 15? (CPO 2022 Pre)

- (a) 999981
- (b) 999951
- (c) 999963
- (d) 999915

152. Find the greatest number which when divides 261, 853 and 1221 leaves a remainder of 5 in each case. (CPO 2023 Pre)

- (a) 19
- (b) 18
- (c) 17
- (d) 16

153. The least number of five digits which is exactly divisible by 9, 12, 15, 25 and 27 is: (CPO 2023 Pre)

- (a) 10800
- (b) 10250
- (c) 10700
- (d) 10600

SSC MTS & SP

154. Let x be the least number which on being divided by 8, 12, 15, 24, 25 and 40 leaves a remainder of 7 in each case. Then find the remainder when x is divided by 19. (MTS 2020 Pre)

- (a) 18
- (b) 27
- (c) 19
- (d) 20

155. Let x be the least number between 70000 and 75000 which on being divided by 225, 250 and 275 leaves a remainder of 61 in each case. The sum of the digits of x is: (MTS 2020 Pre)

- (a) 12
- (b) 29
- (c) 16
- (d) 21

156. Let x be the least number between 56,000 and 60,000 which when divided by 40, 45, 50 and 55 leaves a remainder of 23 in each case. What is the sum of the digits of x ? (MTS 2020 Pre)

- (a) 23
- (b) 21
- (c) 26
- (d) 19

157. When 3820, 4672 and 6163 are divided by the greatest number x , the remainder in each case is the same. What is the quotient when x divided 1035? (MTS 2020 Pre)

- (a) 6
- (b) 8
- (c) 4
- (d) 3

158. What is the smallest number which when divided by 72 and 96, respectively, leaves a remainder 5 in each case? (MTS 2020 Pre)

- (a) 283
- (b) 571
- (c) 581
- (d) 293

159. Which number lying between 800 and 900 is such that when it is divided by either of 34 and 51, the remainder left is 19? (MTS 2020 Pre)

- (a) 887
- (b) 821
- (c) 835
- (d) 869

160. When 5054, 5906 and 7397 are divided by the greatest number x , the remainder in each case is the same. The sum of the digits of x is: (MTS 2020 Pre)

- (a) 11
- (b) 6
- (c) 9
- (d) 5

161. Let x be the smallest 5-digit number such that when it is divided by 5, 6, 7 and 21, it leaves the same remainder 4. What is the sum of the digits of x ? (MTS 2020 Pre)

- (a) 11
- (b) 13
- (c) 10
- (d) 17

SSC MAINS

162. When 7897, 8110 and 8536 are divided by the greatest number x , then the remainder in each case is the same. The sum of the digits of x is: (CGL MAINS 2018)

- (a) 9
- (b) 6
- (c) 5
- (d) 14

163. Let, x be the greatest number which when divides 955, 1027, 1075, the remainder in each case is the same. Which of the following is NOT a factor of x ? (CGL MAINS 2019)

- (a) 6
- (b) 16
- (c) 4
- (d) 8

164. The largest three-digit number that gives the same remainder 2 when divided by 3, 5 and 9 is: (CGL MAINS 2022)

- (a) 999
- (b) 984
- (c) 998
- (d) 992

165. When 860, 1712 and 3203 are divided by a two-digit number x , then the remainder in each case is y . If $(2x + y)$ is written $p \times q \times r^2$, where p , q and r are prime numbers then what is the value of $(p + q + r)$? (IB ACIO-II)

- (a) 12
- (b) 10
- (c) 16
- (d) 15

Largest And Smallest number divisible by No.1, No.2, No.3

SSC CGL

166. What is the sum of the digits of the largest five digit number which is divisible by 5, 35, 39 and 65? (CGL 2020 Pre)

- (a) 33
- (b) 30
- (c) 35
- (d) 27

167. Which of the following is the smallest number that is a perfect square and is divisible by each of the numbers 6, 8 and 15? (CGL 2021 Pre)

- (a) 225
- (b) 121
- (c) 576
- (d) 3600

168. Find the greatest 5-digit number which is divisible by 11, 33, 99 and 121. (CGL 2022 Pre)

- (a) 90099
- (b) 99990
- (c) 99099
- (d) 90909

169. What is the smallest perfect square which is completely divisible by each of 16, 18 and 36? (CGL 2022 Pre)

- (a) 144
- (b) 81
- (c) 196
- (d) 169

170. The largest 5 digit number exactly divisible by 88 is: (CGL 2023 Pre)

- (a) 99990
- (b) 99984
- (c) 99978
- (d) 99968

171. Which of the following is the smallest 5-digits number that is exactly divisible by 526? (CGL 2023 Pre)

- (a) 10520
- (b) 11046
- (c) 10516
- (d) 10426

172. Which of the following is the largest 5-digit number divisible by 47? (CGL 2024 Pre)

- (a) 99999
- (b) 98888
- (c) 99969
- (d) 10000

SSC CPO

173. The greatest number of 5 digits that is exactly divisible by each of 8, 12, 15 and is: (CPO 2018 Pre)

- (a) 99950
- (b) 99940
- (c) 99980
- (d) 99960

174. The greatest number of five digits which is divisible by 13, 15, 18 and 21 is: (CPO 2022 Pre)

- (a) 99120
- (b) 98280
- (c) 96840
- (d) 95830

175. The largest number of four digits that is exactly divisible by 17 and 36 is: (CPO 2023 Pre)

- (a) 7956
- (b) 9180
- (c) 9792
- (d) 8568

176. Which is the smallest natural number that is exactly divisible by each of 96, 108 and 144? (CPO 2023 Pre)

- (a) 864
- (b) 2592
- (c) 1296
- (d) 1728

SSC CHSL

177. The largest five-digit number that is exactly divisible by 81 is: (CHSL 2019 Pre)

- (a) 99989
- (b) 99991
- (c) 99954
- (d) 99876

178. What is the least 5-digit number that is divisible by 91? (CHSL 2019 Pre)

- (a) 10283
- (b) 10101
- (c) 10010
- (d) 10192

179. What is the greatest five-digit number that is completely divisible by 8, 15, 16, 21, and 5? (CHSL 2020 Pre)

- (a) 92680
- (b) 99120
- (c) 95760
- (d) 98320

180. The smallest six-digit number that is exactly divisible by 53 is: (CHSL 2020 Pre)

- (a) 100011
- (b) 100042
- (c) 100008
- (d) 100064

Number System

181. The largest six-digit number exactly divisible by 243 is: (CHSL 2020 Pre)

- (a) 999943
- (b) 999945
- (c) 999947
- (d) 999949

182. Which of the following is the least 6-digit number that is divisible by 93? (CHSL 2021 Pre)

- (a) 100068
- (b) 100070
- (c) 100075
- (d) 100065

183. What is the largest five digit number exactly divisible by 88? (CHSL 2023 Pre)

- (a) 99984
- (b) 99992
- (c) 99986
- (d) 99968

184. What will be the greatest number $32a78b$, which is divisible by 3 but NOT divisible by 9 (Where a and b are single digit number). (CHSL 2023 Pre)

- (a) 329784
- (b) 324781
- (c) 326787
- (d) 329787

SSC MTS & SP

185. The smallest four-digit number which is exactly divisible by 18, 32 and 48 is: (MTS 2020 Pre)

- (a) 1252
- (b) 1156
- (c) 1256
- (d) 1152

186. Let x be the least number divisible by 8, 12, 30, 36 and 45 and x is also a perfect square. What is the value of x ? (MTS 2020 Pre)

- (a) 3600
- (b) 4900
- (c) 2500
- (d) 4225

187. Let x be the greatest number by which when 448, 678 and 908 are divided, the remainder in each case is 11. When 147 is divided by x , the remainder is: (MTS 2020 Pre)

- (a) 4
- (b) 5
- (c) 9
- (d) 3

188. Find the sum of the greatest 4-digit number divisible by 6 and the smallest 4-digit number divisible by 3. (SP Phase-9)

- (a) 11004
- (b) 10998
- (c) 10995
- (d) 11000

189. The smallest six-digit number which is divisible by 12, 16, 24, 28 is: SP 2023 Phase-11 Matric. L.

- (a) 100160
- (b) 100180
- (c) 100190
- (d) 100128

Add/subtract to get perfect square/cube or divisible by n

SSC CGL

190. The least number which should be added to 3627 so that the sum is exactly divisible by 4, 5, 6 and 8 is: (CGL 2021 Pre)
(a) 93 (b) 39
(c) 27 (d) 72
191. Find the smallest number which should be added to the smallest number divisible by 6, 9 and 15 to make it a perfect square. (CGL 2021 Pre)
(a) 10 (b) 9
(c) 19 (d) 21
192. The least number that should be added to 35460 so that the sum is exactly divisible by 3, 4, 5 and 7 is: (CGL 2022 Pre)
(a) 84 (b) 420
(c) 240 (d) 180
193. A six-digit number is divisible by 33. If 54 is added to the number, then the new number formed will also be divisible by: (CGL 2023 Pre)
(a) 3 (b) 2
(c) 5 (d) 7
194. Find the smallest number that can be subtracted from 148109326 so that it becomes divisible by 8. (CGL 2023 Pre)
(a) 4 (b) 8
(c) 6 (d) 10
195. Find the smallest number to be added to 999, so that 99 divides the sum exactly. (CGL 2024 Pre)
(a) 99 (b) 0
(c) 90 (d) 9
196. The least number that should be added to 10000 so that it is exactly divisible by 327 is: (CPO 2018 Pre)
(a) 327 (b) 237
(c) 137 (d) 190
197. The smallest number that should be added to 8212 to obtain a perfect square is: (CPO 2018 Pre)
(a) 123 (b) 69
(c) 54 (d) 112

SSC CPO

198. Which least number should be added to 1000 so that the number obtained is exactly divisible by 37? (CPO 2018 Pre)
(a) 1 (b) 25
(c) 36 (d) 13
199. The least number which should be added to 1351 so that the sum is exactly divisible by 2, 4, 6 and 8 is: (CPO 2023 Pre)
(a) 11 (b) 15
(c) 13 (d) 17
200. What should be subtracted from 246837 to make it divisible by 13? (CHSL 2021 Pre)
(a) 5 (b) 6
(c) 3 (d) 4
201. What is the least number that must be added to the greatest 6-digit number so that sum will be exactly divisible by 294? (CHSL 2023 Pre)
(a) 194 (b) 269
(c) 234 (d) 189
202. What least number must be subtracted from 2963 so that the resulting number when divided by 9, 10 and 15, the remainder in each case is 5? (MTS 2020 Pre)
(a) 41 (b) 78
(c) 82 (d) 39
203. Find the smallest number which should be added to 1234 to get a number divisible by 45. (SP Phase-9)
(a) 16 (b) 8
(c) 26 (d) 1
204. Let, x be the least number which when divided by 15, 18, 20 and 27, the remainder in each case is 10 and x is a multiple of 31. What least number should be added to x to make it a perfect square? (CGL MAINS 2018)
(a) 39 (b) 37
(c) 43 (d) 36
205. Let, x be the least number which when subtracted from 10424 gives a perfect square number. What is the least number by which x should be multiplied to get a perfect square? (CGL MAINS 2019)
(a) 3 (b) 6
(c) 5 (d) 2

SSC MAINS

Number = Quotient × Divisor + Remainder

SSC CGL

206. In a division sum, the divisor is 10 times the quotient and four times the remainder. What is the dividend if the remainder is 45? (CGL 2022 Pre)
(a) 4123 (b) 3285
(c) 2895 (d) 5412
207. In a division sum, the divisor is 13 times the quotient and 6 times the remainder. If the remainder is 39, then the dividend is? (CGL 2023 Pre)
(a) 4240 (b) 4576
(c) 4251 (d) 4800
208. On dividing a number by 38, the quotient is 24 and the remainder is 13, the number is: (CPO 2018 Pre)
(a) 956 (b) 904
(c) 925 (d) 975

SSC CHSL

209. The divisor is 24 times the quotient and 8 times the remainder. If the quotient is 18, then the dividend is: (CHSL 2019 Pre)
(a) 7830 (b) 7630
(c) 7840 (d) 7450
210. On dividing 8675123 by a certain number, the quotient is 33611 and the remainder is 3485. The divisor is: (CHSL 2021 Pre)
(a) 538 (b) 258
(c) 248 (d) 356
211. The divisor is 10 times the quotient and 5 times the remainder in a division sum. What is the dividend if the remainder is 46? (CHSL 2023 Pre)
(a) 5336 (b) 4286
(c) 5972 (d) 4874

SSC MTS & SP

212. Mohan divides 18935 by a certain number. If the quotient and the remainder he gets are 102 and 65, respectively, then the divisor is: (SP 2023 Phase-11 Graduation L)
(a) 185 (b) 155
(c) 165 (d) 175

213. Mohan divides 14817 by a certain number. If the quotient and the remainder he gets are 98 and 117, respectively, then the divisor is: (SP 2023 Phase-11 Graduation L)
(a) 150 (b) 140
(c) 155 (d) 145
214. Two positive numbers differ by 2001. When the larger number is divided by smaller number, the quotient is 9 and the remainder is 41. The sum of the digits of the larger number is: (CGL MAINS 2018)
(a) 15 (b) 11
(c) 10 (d) 14

Remainder

SSC CGL

215. When an integer n is divided by 8, the remainder is 3. What will be the remainder if $6n - 1$ is divided by 8? (CGL 2018 Pre)
(a) 4 (b) 1
(c) 0 (d) 2
216. A integer n is divided by 7 leaves remainder 3. When the 6 times of the number is divided by 7, the remainder is: (CGL 2018 Pre)
(a) 1 (b) 4
(c) 0 (d) 2
217. When 200 is divided by a positive integer x , the remainder is 8. How many values of x are there? (CGL 2019 Pre)
(a) 6 (b) 7
(c) 8 (d) 5
218. When a positive integer divided by d , the remainder is 15. When ten times of the same number is divided by d , the remainder is 6. The least possible value of d is: (CGL 2019 Pre)
(a) 9 (b) 16
(c) 18 (d) 12
219. If 7 divided a positive integer n , the remainder is 2. Which of the following numbers given a remainder of 0 when divided by 7? (CGL 2019 Pre)
(a) $n + 2$ (b) $n + 5$
(c) $n - 5$ (d) $n + 1$
220. What is the remainder when we divide $5^{70} + 7^{70}$ by 74? (CGL 2019 Pre)
(a) 5 (b) 0
(c) 7 (d) 1

221. What is the remainder when the product of 335, 608 and 853 is divided by 13? (CGL 2021 Pre)
(a) 11 (b) 12
(c) 6 (d) 7
222. Which is the smallest multiple of 7, which leaves 5 as remainder in each case, when divided by 8, 9, 12 and 15? (CGL 2021 Pre)
(a) 365 (b) 1085
(c) 2525 (d) 725
223. If each of the two numbers 5^{16} and 5^{25} are divided by 6, the remainders are R_1 and R_2 respectively. What is the value of $\frac{R_1 + R_2}{R_2}$? (CGL 2021 Pre)
(a) $\frac{1}{6}$ (b) $\frac{5}{6}$
(c) $\frac{1}{5}$ (d) $\frac{5}{5}$
224. If 8A5146B is divisible by 88, then what is the value of A^B ? (CGL 2021 Pre)
(a) 27 (b) 64
(c) 81 (d) 12
225. What will be the remainder when 7^{42} is divided by 48? (CGL 2022 Pre)
(a) 1 (b) 3
(c) 2 (d) 0
226. A number n when divided by 6, leaves a remainder 3. What will be the remainder when $(n^2 + 5n + 8)$ is divided by 6? (CGL 2022 Pre)
(a) 1 (b) 3
(c) 5 (d) 2
227. A number when divided by 221, leaves a remainder 30. If the same number is divided by 13, the remainder will be: (CGL 2022 Pre)
(a) 4 (b) 3
(c) 2 (d) 1
228. When $m^{12} - 1$ is divided by $m + 1$, the remainder is: (CGL 2022 Pre)
(a) 2 (b) 1
(c) 0 (d) -1
229. What will be the remainder when $(265)^{6001} + 9$ is divided by 266? (CGL 2023 Pre)
(a) 8 (b) 6
(c) 1 (d) 9
230. What is the remainder when $(x^{12} + 1)$ is divided by $(x + 1)$? (CGL 2023 Pre)
(a) x (b) $x - 1$
(c) 0 (d) 1

231. Find the remainder: $(17^{13} - 21) + 18$ (CGL 2024 Pre)
(a) 15 (b) 17
(c) 14 (d) 21
232. If $(77^{77} + 77)$ is divided by 78, then the remainder will be: (CGL 2024 Pre)
(a) 76 (b) 77
(c) 78 (d) 1
233. Find the remainder when $9^{30} + 2$ is divided by 4. (CGL 2024 Pre)
(a) 1 (b) 3
(c) 0 (d) 2
234. A number r when divided by 8 leaves remainder 3. What will be the remainder when $(r^2 + 6r + 7)$ is divided by 8? (CGL 2024 Pre)
(a) 2 (b) 3
(c) 1 (d) 4
235. Find the remainder, if 19^{200} is divided by 20. (CGL 2024 Pre)
(a) 2 (b) -1
(c) 1 (d) 3
236. On dividing a number by 55, we get 28 as the remainder. On dividing the same number by 11, what is the remainder? (CGL 2024 Pre)
(a) 5 (b) 8
(c) 7 (d) 6
237. Without doing the actual division, find the remainder when 28735429 is divided by 9. (CGL 2024 Pre)
(a) 4 (b) 2
(c) 8 (d) 9

SSC CPO

238. When the integer n is divided by 7, the remainder is 3. What is the remainder if $5n$ is divided by 7? (CPO 2018 Pre)
(a) 0 (b) 3
(c) 2 (d) 1
239. When 2388, 4309 and 8151 are divided by a certain 3-digit number, the remainder in each is the same. The remainder is: (CPO 2019 Pre)
(a) 39 (b) 23
(c) 19 (d) 15
240. Let x be the least number divisible by 16, 24, 30, 36 and 45, and x is also a perfect square. What is the remainder when x is divided by 123? (CPO 2019 Pre)
(a) 100 (b) 40
(c) 103 (d) 33

241. When a certain number is divided by 65, the remainder is 56. When the same number is divided by 13, the remainder is x . What is the value of

$$\sqrt{5x-2}?$$

- (a) $2\sqrt{7}$ (b) $\sqrt{13}$
(c) $2\sqrt{2}$ (d) $3\sqrt{2}$

242. When 3738, 5659 and 9501 are divided by the greatest possible number x , the remainder in each case is y . What is the sum of x and y ? (CPO 2019 Pre)

- (a) 3738 (b) 3783
(c) 3673 (d) 3637

243. If $14331433 \times 1422 \times 1425$ is divided by 12, then what is the remainder? (CPO 2020 Pre)

- (a) 8 (b) 3
(c) 9 (d) 6

244. The remainder when $72 \times 73 \times 78 \times 76$ is divided by 35 is:

$$(CPO 2020 Pre)$$

- (a) 15 (b) 8
(c) 22 (d) 12

SSC CHSL

245. Given n is an integer, what is the remainder when $(6n + 3)^2$ is divided by 9? (CHSL 2018 Pre)

- (a) 3 (b) 2
(c) 1 (d) 0

246. When $(77^{77} + 77)$ is divided by 78, the remainder is:

$$(CHSL 2019 Pre)$$

- (a) 75 (b) 77
(c) 76 (d) 74

If $2x^2 + ax^2 + bx - 2$ leaves the remainders 7 and 0 when divided by $(2x - 3)$ and $(x + 2)$, respectively, then the values of a and b are respectively.

$$(CHSL 2019 Pre)$$

- (a) 3; -3 (b) 2; -2
(c) -3; 3 (d) -2; 2

248. If a number is divided by 3, the remainder will be 2. If the number is added by 5 and then divided by 3, then what will be the remainder? (CHSL 2019 Pre)

- (a) 3 (b) 1
(c) 2 (d) 0

249. If a positive integer n is divided by 7, the remainder is 2. Which of the number in the options yields a remainder of 0 when it is divided by 7?

$$(CHSL 2019 Pre)$$

- (a) $n + 3$ (b) $n + 1$
(c) $n + 2$ (d) $n + 5$

250. If $kx^3 + 4x^2 + 3x - 4$ and $x^3 - 4x + k$ leave the same remainder when divided by $(x - 3)$, then the value of k is:

$$(CHSL 2019 Pre)$$

- (a) 2 (b) -1
(c) 1 (d) 0

251. When a number is divided by 14, the remainder is 9. If the square of the same number is divided by 14, then the remainder will be:

$$(CHSL 2019 Pre)$$

- (a) 11 (b) 9
(c) 10 (d) 8

252. If a number is divided by 899, then remainder is 63. If the same number is divided by 29, then remainder will be:

$$(CHSL 2019 Pre)$$

- (a) 10 (b) 2
(c) 4 (d) 5

253. When $(2^{24} - 1)$ is divided by 7, the remainder is:

$$(CHSL 2020 Pre)$$

- (a) 2 (b) 0
(c) 4 (d) 1

254. If a number is divisible by 624, the remainder will be 53. If the same number is divisible by 16, then the remainder will be:

$$(CHSL 2020 Pre)$$

- (a) 7 (b) 6
(c) 4 (d) 5

255. When an integer n is divided by 6, the remainder is 5. What is the remainder if $9n$ is divided by 6? (CHSL 2020 Pre)

- (a) 3 (b) 2
(c) 4 (d) 5

256. When a number M is divided by 7, the remainder is 6. What is the remainder if the square of M is divided by 7?

$$(CHSL 2020 Pre)$$

- (a) 4 (b) 1
(c) 3 (d) 2

257. When $f(x) = 15x^3 - 14x^2 - 4x + 10$ is divided by $(3x + 2)$, then the remainder is:

$$(CHSL 2021 Pre)$$

- (a) -1 (b) 1
(c) -2 (d) 2

258. If a positive integer N is divided by 7, the remainder is 3. Which of the following numbers yields a remainder of 0 when it is divided by 7? (CHSL 2021 Pre)

- (a) $N + 5$ (b) $N + 2$
(c) $N + 4$ (d) $N + 3$

259. When two numbers are separately divided by 44, the remainders are 11 and 38, respectively. If the sum of the two numbers is divided by 44, then the remainder will be:

$$(CHSL 2021 Pre)$$

- (a) 16 (b) 4
(c) 33 (d) 5

260. A number when divided by 78 gives the quotient 280 and the remainder 0. If the same number is divided by 65, what will be the value of the remainder? (CHSL 2023 Pre)

- (a) 0 (b) 2
(c) 1 (d) 3

261. What is the remainder when 3^8 is divided by 7?

$$(CHSL 2023 Pre)$$

- (a) 4 (b) 2
(c) 5 (d) 6

262. Two numbers, when divided by a certain divisor, leave the remainder 57. When sum of the two numbers is divided by the same divisor, the remainder is 49. The divisor is:

$$(CHSL 2023 Pre)$$

- (a) 56 (b) 57
(c) 49 (d) 65

263. If 7 divides the integer n , then the remainder is 2. What will be the remainder if $9n$ is divided by 7? (CHSL 2023 Pre)

- (a) 5 (b) 3
(c) 1 (d) 4

264. The remainder of the term $9^1 + 9^2 + \dots + 9^{2n+1}$ when divided by 6 is:

$$(CHSL 2023 Pre)$$

- (a) 2 (b) 1
(c) 4 (d) 3

265. A natural number n divides 732 and leaves 12 as a remainder. How many values of n are possible? (CHSL 2024 Pre)

- (a) 18 (b) 20
(c) 16 (d) 22

SSC MTS & SP

266. What is the remainder when we divide $4^{50} + 7^{50}$ by 65?

$$(SP Phase-8)$$

- (a) 3 (b) 2
(c) 0 (d) 1

267. If $14331433 \times 1423 \times 1425$ is divided by 12, then what will be the remainder? (SP Phase-8)

- (a) 2 (b) 3
(c) 1 (d) 0

268. When a certain number is divided by 52, the remainder is 49. When the same number is divided by 13, the remainder is x . What is the value of $\sqrt{5x-1}$?

$$(SP Phase-8)$$

- (a) 6 (b) 7
(c) 8 (d) 11

269. When 200 is divided by a positive integer x . The remainder is 12. How many values of x are possible?

$$(SP Phase-8)$$

- (a) 3 (b) 6
(c) 2 (d) 8

270. When number n is divided by 6, the remainder is 3. What will be the remainder when $(n^4 + n^3 + n^2 + 5n)$ is divided by 6?

$$(SP Phase-9)$$

- (a) 2 (b) 1
(c) 0 (d) 3

271. When $(2^{18} - 1)$ is divided by 9, the remainder is:

$$(SP Phase-9)$$

- (a) 2 (b) 0
(c) 8 (d) 1

SSC MAINS

272. If x is the remainder when 3^{61284} is divided by 5 and y is the remainder when 4^{36} is divided by 6, then what is the value of $(2x - y)$? (CGL MAINS 2018)

- (a) -4 (b) 4
(c) -2 (d) 2

273. What is the remainder when $(127^{97} + 97^{97})$ is divided by 32?

$$(CGL MAINS 2018)$$

- (a) 4 (b) 2
(c) 7 (d) 0

274. When positive numbers a , b and c are divided by 13, the remainders are 9, 7 and 10 respectively. What will be the remainder when $(a + 2b + 5c)$ is divided by 13? (CGL MAINS 2019)

- (a) 10 (b) 5
(c) 9 (d) 8

Largest and smallest number divisible by No.1, No.2, No.3 Leaves remainder x_1, x_2, x_3

SSC CGL

275. The greatest number that divides 126, 224 and 608 leaving remainders 2, 7 and 19, respectively, is: (CGL 2021 Pre)

- (a) 27 (b) 31
(c) 21 (d) 37

276. Which is the largest six digit number, which when divided by 12, 15, 20, 24 and 30, leaves the remainders 8, 11, 16, 20 and 26 respectively.

$$(CPO 2019 Pre)$$

- (a) 999956 (b) 999960
(c) 999964 (d) 999982

277. Which of the following numbers leaves the remainder equal to the highest common factor of 6, 8 and 9, when divided by 6, 8 and 9?

$$(CPO 2023 Pre)$$

- (a) 291 (b) 575
(c) 506 (d) 433

SSC CHSL

278. Find the number when successively divided by 3, 5 and 7 leaves remainder 2, 1 and 3, respectively, and the last quotient is 3. (CHSL 2024 Pre)

- (a) 360 (b) 362
(c) 365 (d) 367

279. After the division of a number successively by 2, 3 and 5, the remainders are 1, 2 and 3, respectively. What will be the remainder, if 13 divides the same number (if the last quotient is 1)? (CHSL 2024 Pre)

- (a) 1 (b) 2
(c) 0 (d) 3

SSC MTS & SP

280. Which is the largest number that will divide 1992 and 233 leaving remainders 12 and 13, respectively? (MTS 2020 Pre)

- (a) 242 (b) 220
(c) 246 (d) 186

281. Let x be the smallest 4-digit number such that when it is divided by 5, 6 and 7, it leaves the remainders 2, 3 and 4, respectively. The sum of the digits of x is: (MTS 2020 Pre)

- (a) 9 (b) 11
(c) 12 (d) 14

SSC MAINS

282. The sum of the digits of the least number which when divided by 36, 72, 80 and 88 leaves the remainders 16, 52, 60 and 68, respectively, is:

$$(CGL MAINS 2019)$$

- (a) 17 (b) 11
(c) 14 (d) 16

Counting of numbers divisible by n

SSC CGL

283. How many numbers are there from 200 to 800 which are neither divisible by 5 nor by 7?

$$(CGL 2019 Pre)$$

- (a) 407 (b) 410
(c) 413 (d) 411

284. How many numbers are there from 500 to 650 (including both) which are neither divisible by 3 nor by 7? (CGL 2021 Pre)

- (a) 21 (b) 121
(c) 87 (d) 99

285. What is the sum of the numbers between 400 and 500 such that when they are divided by 6, 12 and 16, it leaves no remainder? (CGL 2021 Pre)

- (a) 40 (b) 1024
(c) 960 (d) 912

SSC CPO

286. How many natural numbers less than 1000 are divisible by 5 or 7 but NOT by 35?

$$(CPO 2019 Pre)$$

- (a) 285 (b) 313
(c) 341 (d) 243

287. How many natural numbers up to 2001 are divisible by 3 or 4 but NOT by 5? (CPO 2019 Pre)

- (a) 768 (b) 801
(c) 934 (d) 1067

288. How many numbers between 400 and 700 are divisible by 5, 6 and 7? (CPO 2020 Pre)

- (a) 20 (b) 10
(c) 2 (d) 5

289. How many numbers between 300 and 700 are divisible by 5, 6 and 8? (CPO 2020 Pre)

- (a) 20 (b) 5
(c) 2 (d) 3

290. Ramu had to select a list of numbers between 1 and 1000 (including both), which are divisible by both 2 and 7. How many such numbers are there? (CPO 2022 Pre)

- (a) 142 (b) 71
(c) 97 (d) 642

SSC CHSL

291. How many two-digit numbers are divisible by 3 or 5?

(CHSL 2021 Pre)

- (a) 48 (b) 41
(c) 42 (d) 40

292. How many numbers are there between 201 and 401 which are divisible by 5 but not by 4?

(CHSL 2022 Pre)

- (a) 30 (b) 20
(c) 40 (d) 45

293. What is the sum of all two digit numbers which are divisible by 9?

(CHSL 2022 Pre)

- (a) 565 (b) 535
(c) 585 (d) 475

294. How many numbers are there between 501 and 701 which are divisible by 10 but are not divisible by 3?

(CHSL 2022 Pre)

- (a) 20 (b) 14
(c) 15 (d) 13

295. How many numbers are there between 17 and 457 which are divisible by both 5 and 3?

(CHSL 2022 Pre)

- (a) 33 (b) 29
(c) 35 (d) 30

296. What is the sum of all three digit numbers which are divisible by 15?

(CHSL 2022 Pre)

- (a) 41200 (b) 36825
(c) 32850 (d) 28750

SSC MTS & SP

297. How many numbers between 500 and 900, both inclusive, are exactly divisible by all the numbers, 12, 15, 20 and 30?

(MTS 2020 Pre)

- (a) 4 (b) 5
(c) 7 (d) 6

298. What is the sum of the number between 300 and 500 such that when they are divided by 6, 12 and 16, it leaves no remainder?

(MTS 2020 Pre)

- (a) 1586 (b) 1632
(c) 1764 (d) 1618

299. How many integers between 299 and 501 are divisible by 4 or 10?

(SP Phase-9)

- (a) 56 (b) 72
(c) 51 (d) 61

300. How many numbers are there from 500 to 550 (including both) which are neither divisible by 3 nor by 11?

(SP Phase-9)

- (a) 18 (b) 11
(c) 28 (d) 30

301. The 6-digit number $439xy5$ is divisible by 125. How many such 6-digit numbers are there?

(SP Phase-9)

- (a) 5 (b) 2
(c) 4 (d) 3

SSC MAINS

302. How many numbers are there from 400 to 700 in which the digit 6 occurs exactly twice?

(CGL MAINS 2021)

- (a) 19 (b) 18
(c) 21 (d) 20

303. How many composite numbers are there from 53 to 97?

(CGL MAINS 2021)

- (a) 36 (b) 38
(c) 37 (d) 35

304. How many whole numbers lie between 11^2 and 12^2 ?

(CHSL 2022 Mains)

- (a) 23 (b) 24
(c) 21 (d) 22

Number of factors

305. What is the sum of the divisors of 484 that are perfect squares?

(CGL 2023 Pre)

- (a) 125 (b) 35
(c) 610 (d) 13

306. The number of factors of 196 which are divisible by 4 is:

(CPO 2022 Pre)

- (a) 228 (b) 4
(c) 3 (d) 57

307. The sum of the odd divisors of 216 is:

(CPO 2022 Pre)

- (a) 16 (b) 14
(c) 40 (d) 600

SSC CHSL

308. What is the total number of factors of the number 840 except 1 and the number itself?

(CHSL 2022 Pre)

- (a) 29 (b) 30
(c) 28 (d) 31

309. Which of the following sets is such that all its elements are divisors of the number 2520?

(CHSL 2023 Pre)

- (a) 12, 49, 18 (b) 16, 15, 14
(c) 21, 10, 25 (d) 8, 9, 7

310. How many factors of 14,400 are divisible by 18 but not by 36?

(CHSL 2024 Pre)

- (a) 5 (b) 2
(c) 4 (d) 3

311. How many distinct natural numbers are the divisors of $(30)^{12}$?

(SP Phase-8)

- (a) 124 (b) 122
(c) 123 (d) 125

SSC MAINS

312. The number of factors of 3600 is:

(CGL MAINS 2018)

- (a) 45 (b) 44
(c) 43 (d) 42

313. Find the number of prime factors in the product $(30)^5 \times (24)^5$.

(CGL MAINS 2019)

- (a) 45 (b) 10
(c) 35 (d) 30

314. The smallest prime number that is a divisor of 17408 is 2 ($17408 = 17 \times 2^{10}$). What is the sum of digits of the greatest prime number, which is a divisor of 17391?

(IB ACIO-III)

- (a) 5 (b) 6
(c) 8 (d) 4

Leaves same remainder when divisible by n

SSC CGL

315. What is the sum of digits of the least number. Which when divided by 15, 18 and 24 leaves the remainder 8 in each case and is also divisible by 13?

(CPO 2018 Pre)

- (a) 18 (b) 16
(c) 15 (d) 17

SSC CPO

316. What is the sum of digits of the least number, which when divided by 15, 18 and 42 leaves the same remainder 8 in each case and is also divisible by 13?

(CPO 2018 Pre)

- (a) 24 (b) 26
(c) 25 (d) 22

317. Let x be the least number which when divided by 8, 9, 12, 14 and 36 leaves a remainder of 4 in each case, but x is divisible by 11. The sum of the digits of x is

(CPO 2019 Pre)

- (a) 9 (b) 4
(c) 6 (d) 5

SSC MAINS

318. When 12, 16, 18, 20 and 25 divide the least number x, the remainder in each case is 4 but x divisible by 7. What is the digit at the thousands place in x?

(CGL MAINS 2018)

- (a) 5 (b) 4
(c) 8 (d) 3

319. Find the least number which when divided by 12, 18, 24 and 30 leaves 4 as remainder in each case, but when divided by 7 leaves no remainder.

(CGL MAINS 2019)

- (a) 366 (b) 634
(c) 384 (d) 364

Successive division

320. When a number is successively divided by 3, 4 and 7, the remainders obtained are 2, 3 and 5, respectively. What will be the remainder when 84 divides the same number?

(CPO 2020 Pre)

- (a) 71 (b) 30
(c) 48 (d) 53

SSC CHSL

321. When a number is divided by 3, the remainder is 2. Again, when the quotient is divided by 7, the remainder is 5. What will be the remainder when the original number is divided by 21?

(CHSL 2020 Pre)

- (a) 14 (b) 13
(c) 17 (d) 16

322. Successive division of a number by 2, 3, 5 and 7 gives remainders 1, 4, 0 and 5, respectively. What will be the sum of the remainders if the same number is divided by 7, 5, 3 and 2 successively?

(CHSL 2024 Pre)

- (a) 8 (b) 9
(c) 10 (d) 17

323. When positive numbers x, y and z are divided by 31, the remainders are 17, 24 and 27, respectively. When $(4x - 2y + 3z)$ is divided by 31, the remainder will be:

(CGL MAINS 2019)

- (a) 9 (b) 8
(c) 16 (d) 19

Series

SSC MAINS

$$324. \frac{1}{15} + \frac{1}{35} + \frac{1}{63} + \frac{1}{99} + \frac{1}{143} = ?$$

(CGL MAINS 2019)

- (a) $\frac{5}{39}$ (b) $\frac{7}{39}$

- (c) $\frac{4}{39}$ (d) $\frac{10}{39}$

325. Find the sum of $6 + 8 + 10 + 12 + 14 + \dots + 40$.

(CGL MAINS 2019)

- (a) 424 (b) 1600
(c) 400 (d) 414

$$326. \frac{3}{70} + \frac{1}{42} + \frac{1}{66} + \frac{3}{286} + \frac{1}{130} + \frac{1}{170} = ?$$

(CGL MAINS 2019)

- (a) $\frac{7}{70}$ (b) $\frac{3}{85}$

- (c) $\frac{11}{85}$ (d) $\frac{9}{85}$

327. What is the value of $99\frac{11}{99} + 99\frac{13}{99} + 99\frac{15}{99} + \dots + 99\frac{67}{99}$?

(CGL MAINS 2021)

- (a) 94220/33 (b) 95120/33
(c) 97120/33 (d) 96220/33

328. What is the sum of all the common terms between the given series S1 and S2?
S1 = 2, 9, 16, ..., 632
S2 = 7, 11, 15, ..., 743

(CGL MAINS 2021)

- (a) 6974 (b) 6750
(c) 7140 (d) 6860

329. What is the sum of first 20 terms of the following series?
 $1 \times 2 + 2 \times 3 + 3 \times 4 + 4 \times 5 + \dots$

(CGL MAINS 2021)

- (a) 3160 (b) 2940
(c) 3240 (d) 3080

330. What is the value of

$$\frac{7}{2} + \frac{11}{3} + \frac{7}{6} + \frac{11}{15} + \frac{7}{12} + \dots$$

$$\frac{11}{35} + \dots + \frac{7}{156} + \frac{11}{575} = ?$$

(CGL MAINS 2021)

- (a) 3917/355 (b) 3816/325
(c) 3714/345 (d) 3216/315

Recurring Decimal

SSC MAINS

$$331. 22.\bar{4} + 11.5\bar{67} - 33.5\bar{9} = ?$$

(CGL MAINS 2018)

- (a) $0.4\bar{12}$ (b) $0.3\bar{1}$
(c) $0.4\bar{12}$ (d) $0.3\bar{2}$

332. The value of $0.5\bar{6} - 0.7\bar{23} + 0.3\bar{9}$

$\times 0.7$ is:

(CGL MAINS 2018)

- (a) $0.1\bar{54}$ (b) $0.1\bar{54}$
(c) $0.1\bar{58}$ (d) $0.1\bar{58}$

333. $0.4\bar{7} + 0.5\bar{03} - 0.3\bar{9} \times 0.8 = ?$

(CGL MAINS 2018)

- (a) $0.6\bar{15}$ (b) $0.6\bar{15}$
(c) $0.6\bar{25}$ (d) $0.6\bar{25}$

334. The value of $(2.\bar{4} \times 0.\bar{6} \times 3 \times 0.1\bar{6})$

$\times [0.2\bar{7} \times (0.8\bar{3} + 0.1\bar{6})]$ is:

(CGL MAINS 2019)

- (a) $0.8\bar{14}$ (b) $0.1\bar{1}$
(c) $1.\bar{1}$ (d) $1.3\bar{6}$

335. The value of $0.4\bar{6} + 0.7\bar{23} - 0.3\bar{9} \times 0.7$ is:

(CGL MAINS 2019)

- (a) $0.9\bar{7}$ (b) $0.8\bar{7}$
(c) $0.5\bar{7}$ (d) $0.7\bar{7}$

336. The value of $11.\bar{4} + 22.5\bar{67} - 33.5\bar{9}$

- (a) $40.\bar{12}$ (b) $4.\bar{12}$
(c) $0.4\bar{12}$ (d) $0.04\bar{12}$

337. If $A = 0.3\bar{12}$, $B = 0.4\bar{15}$ and $C = 0.30\bar{9}$, then what is the value of $A + B + C$?

(CGL MAINS 2021)

- (a) $1141/1100$ (b) $1097/1100$
(c) $1211/1100$ (d) $1043/1100$

Ladder Fractions

SSC MAINS

$$338. \text{If } \frac{45}{53} = \frac{1}{a + \frac{1}{b + \frac{1}{c - \frac{2}{5}}}}, \text{ where a, b}$$

and c are positive integers, then what is the value of $(4a - b + 3c)$?

(CGL MAINS 2019)

- (a) 5 (b) 4
(c) 6 (d) 7

$$339. \text{ If } \frac{1}{x} + \frac{1}{y} = \frac{29}{79} \text{ where } x, y \text{ and } z \text{ are natural numbers, then the value of } (2x + 3y - z) \text{ is:}$$

- (a) 1 (b) 4
(c) 0 (d) 2

$$340. \text{ If } 2 = x + \frac{1}{1 + \frac{1}{5 + \frac{1}{2}}}, \text{ then the value of } x \text{ is equal to:}$$

- (a) 15/13 (b) 14/13
(c) 13/15 (d) 1

Unit Digit

$$341. \text{ If } N = (307)^{35} + (524)^{20}, \text{ then what is the unit digit of } N?$$

- (a) 6 (b) 5
(c) 3 (d) 4

$$342. \text{ Find the unit digit in } (14)^{112} + (14)^{113}.$$

- (a) 2 (b) 8
(c) 0 (d) 4

SSC MAINS

$$343. \text{ If } x = (633)^{15} - (277)^{35} + (266)^{14}, \text{ what is the unit digit of } x?$$

- (a) 4 (b) 7
(c) 6 (d) 8

$$\text{If } x = (164)^{16} + (333)^{17} - (727)^{20}, \text{ then what is the unit digit of } x?$$

- (a) 5 (b) 7
(c) 8 (d) 9

$$345. \text{ Let, } x = (433)^{24} - (377)^{36} + (166)^{34}. \text{ What is the unit digit of } x?$$

- (a) 8 (b) 7
(c) 6 (d) 9

Number Formation

SSC MTS & SP

$$346. \text{ A six-digit number 4ABB8A is a multiple of 33 for non-zero digits A and B. Which of the following could be the possible values of A and B? (SP Phase-8)}$$

- (a) A = 3, B = 2 (b) A = 6, B = 2
(c) A = 3, B = 6 (d) A = 6, B = 3

$$347. \text{ Which among the following can be expressed as } (10p+q)(10q+p), \text{ where p and q are integers?}$$

- (a) 1456 (b) 1205
(c) 1729 (d) 1500

$$348. \text{ Find the smallest 4-digit number divisible by 8, formed using the digits 0, 2, 3, 4, 6, 8 without any repetition. (SP Phase-9)}$$

- (a) 2048 (b) 2408
(c) 3024 (d) 2034

$$349. \text{ Find the greatest 3-digit number divisible by 3, formed using the digits 0, 1, 2, 3, 5 without any repetition. (SP Phase-9)}$$

- (a) 513 (b) 510
(c) 531 (d) 532

SSC MAINS

$$350. \text{ The sum of the digits of a two-digit number is } 1/7 \text{ of the number. The units digit is 4 less than the tens digit. If the number obtained on reversing its digits is divided by 7, the remainder will be:}$$

- (a) 1 (b) 5
(c) 6 (d) 4

$$351. \text{ When a two-digit number is multiplied by the sum of its digits, the product is 424. When the number obtained by interchanging its digits is multiplied by the sum of the digits, the result is 280. The sum of the digits of the given number is:}$$

- (a) 6 (b) 9
(c) 8 (d) 7

$$352. \text{ The denominator of a fraction is 4 more than twice the numerator. When the numerator is increased by 3 and the denominator is decreased by 3, the fraction becomes } \frac{2}{3}. \text{ What is the difference between the denominator and numerator of the original fraction?}$$

- (a) 13 (b) 11
(c) 12 (d) 10

$$353. \text{ Let ab, a \neq b \text{ be 2-digit prime number such that ba is also a prime number. The sum of all such number is: (CGL MAINS 2019)}$$

- (a) 374 (b) 418
(c) 407 (d) 396

Number System

$$354. \text{ The numerator of a fraction is 6 less than its denominator. If 1 is subtracted from its numerator and 5 is added to its denominator, then its denominator becomes 4 times its numerator. Find the fraction. (CGL MAINS 2019)}$$

- (a) $\frac{4}{11}$ (b) $\frac{5}{11}$
(c) $\frac{7}{11}$ (d) $\frac{3}{11}$

$$355. \text{ In a two-digit number, its units digit exceeds its tens digit by 2 and that the product of the given number and the sum of its digits is equal to 460. The number is: (CGL MAINS 2019)}$$

- (a) 48 (b) 64
(c) 36 (d) 46

$$356. \text{ If the sum of two positive numbers is 65 and the square root of their product is 26, then the sum of their reciprocals is: (CGL MAINS 2019)}$$

- (a) $\frac{3}{52}$ (b) $\frac{7}{52}$
(c) $\frac{1}{52}$ (d) $\frac{5}{52}$

$$357. \text{ Let, p, q, r and s be positive natural numbers having three exact factors including 1 and the number itself. If } q > p \text{ and both are two-digit numbers, and } r > s \text{ and both are one-digit numbers, then the value of the expression } \frac{p-q-1}{r-s} \text{ is:}$$

- (a) -S - 1 (b) S - 1
(c) 1 - S (d) S + 1

$$358. \text{ If the digits of a two digit number is reversed, then the number is decreased by 36. Which of the following is correct regarding the number? I. The difference of the digits is 4.$$

- II. The value of number can be 84.

$$III. \text{ Number is always a composite number. (CGL MAINS 2021)}$$

- (a) I, II and III
(b) II and III
(c) I and III
(d) I and II

$$359. \text{ Let X be a two digit number and Y be another two digit number formed by interchanging the digits of X. If } (X + Y) \text{ is the greatest two digit number, then what is the number of possible value of X? (UPSC CSAT 2024)}$$

- (a) 2 (b) 4
(c) 6 (d) 8

$$360. \text{ If the sum of the two digit numbers AB and CD is the three digit number 1CE, where the letters A, B, C, D, E denote distinct digits, then what is the value of A? (UPSC CSAT 2024)}$$

- (a) 9 (b) 8
(c) 7 (d) CND

Miscellaneous

SSC CGL

$$361. \text{ The sum of the squares of 3 natural numbers is 1029, and they are in the proportion 1:2:4. The difference between greatest number and smallest number is: (CGL 2019 Pre)}$$

- (a) 21 (b) 18
(c) 15 (d) 31

$$362. 25a^2 - 9 \text{ is factored as: (CGL 2019 Pre)}$$

- (a) $(5a + 3)(5a - 3)$
(b) $(5a - 3)^2$
(c) $(25a + 1)(a - 9)$
(d) $(5a + 1)(5a - 9)$

$$363. \text{ If a number P is divisible by 2 and another number Q is divisible by 3, then which of the following is true? (CGL 2020 Pre)}$$

- (a) P × Q is divisible by 6
(b) P + Q is divisible by 6
(c) P + Q is divisible by 5
(d) P × Q is divisible by 5

$$364. \text{ If } (56\sqrt{7}x^3 - 2\sqrt{2}y^3) + (2\sqrt{7}x - \sqrt{2}y) = Ax^2 + By^2 - Cxy, \text{ then find the value of } A + B - \sqrt{14}C. \text{ (CGL 2020 Pre)}$$

- (a) 19 (b) 10
(c) 58 (d) 38

$$365. \text{ In an examination, the average score of a student was 67.6. If he would have got 27 more marks in Mathematics, 10 more marks in Computer Science, 18 more marks in History and retained the same marks in other subjects, then his average score would have been 72.6. How many papers were there in the examination? (CGL 2020 Pre)}$$

- (a) 11 (b) 10
(c) 12 (d) 9

$$366. \text{ What is the average of the first six prime numbers? (CGL 2021 Pre)}$$

- (a) 6 (b) $6\frac{5}{6}$
(c) $9\frac{1}{3}$ (d) 7

$$367. \text{ If } \frac{(17)^3 + (7)^3}{(17^2 + 7^2 - k)} = 24, \text{ then what is the value of k? (CGL 2022 Pre)}$$

- (a) 119 (b) 128
(c) 24 (d) 109

$$368. \text{ Simplify the following (CGL 2022 Pre)}$$

- $25^3 - 75^3 + 50^3$
(a) -281250 (b) 281350
(c) 271250 (d) -281450

$$369. \text{ The value of } 97 \times 103 \text{ is (CGL 2022 Pre)}$$

- (a) 7999 (b) 9991
(c) 8991 (d) 9981

$$370. 7 \text{ is added to a certain number and the sum is multiplied by 5. The product is then divided by 3 and 4 is subtracted from the quotient. If the result comes to 16, then what is the original number? (CGL 2022 Pre)}$$

- (a) 3 (b) 1
(c) 5 (d) 4

$$371. \text{ Choose the option in which the numbers are in correct ascending order. (CGL 2022 Pre)}$$

- (a) $\frac{4}{5}, \frac{2}{3}, \frac{1}{11}$ and $\frac{2}{9}$
(b) $\frac{1}{11}, \frac{2}{9}, \frac{2}{5}$ and $\frac{4}{5}$
(c) $\frac{2}{9}, \frac{1}{11}, \frac{4}{5}$ and $\frac{2}{3}$
(d) $\frac{2}{3}, \frac{4}{5}, \frac{1}{11}$ and $\frac{2}{9}$

$$372. \text{ Find the weighted arithmetic mean of the first 'n' natural numbers, the weights being the corresponding numbers. (CGL 2022 Pre)}$$

- (a) $\frac{n(n+1)(2n+1)}{2}$
(b) $\frac{n(n+1)}{2}$
(c) n (d) $\frac{2n+1}{3}$

$$373. \text{ What is the value of } 100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + 94^2 - 93^2 + \dots + 12^2 - 11^2? \text{ (CGL 2022 Pre)}$$

- (a) 5050 (b) 4985
(c) 4995 (d) 4950

Number System

$$374. \text{ Which of the following numbers is a divisor of } (49^{15} - 1)? \text{ (CGL 2022 Pre)}$$

- (a) 46 (b) 14
(c) 8 (d) 50

$$375. \text{ In a class of students, the first student has 2 toffees, second has 4 toffees, third has 6 toffees and so on. If the number of students in the class is 25, then the total number of toffees are divisible by (CGL 2022 Pre)}$$

- (a) 5 and 7 (b) 5 and 13
(c) 11 and 13 (d) 7 and 11

$$376. \text{ The difference of two numbers is 1564. After dividing the larger number by the smaller, we get 6 as quotient and 19 as remainder. What is the smaller number? (CGL 2022 Pre)}$$

- (a) 456 (b) 287
(c) 623 (d) 309

$$377. \text{ A four-digit pin, say abcd, of a lock has different non-zero digits. The digits satisfy } b = 2a, c = 2b, d = 2c. \text{ The pin is divisible by (CGL 2022 Pre)}$$

- (a) 2, 3, 5 (b) 2, 3, 7
(c) 2, 3, 13 (d) 2, 3, 11

$$378. \text{ The cube of the difference between two given natural numbers is 1728, while the product of these two given numbers is 108. Find the positive difference between the cubes of these two given numbers? (CGL 2023 Pre)}$$

- (a) 4104 (b) 5616
(c) 2160 (d) 5626

$$379. \text{ The square of the sum of two given natural numbers is 784, while the product of the two given numbers is 192. Find the positive difference between the squares of these two given numbers. (CGL 2023 Pre)}$$

- (a) 512 (b) 122
(c) 400 (d) 112

$$380. \text{ The difference between the cubes of two given natural numbers is 6272, while the positive difference between the two given numbers is 8. What is the sum of the cubes of the two given numbers? (CGL 2023 Pre)}$$

- (a) 9728 (b) 9684
(c) 8000 (d) 9600

381. The sum of the cubes of two given natural numbers is 9728, while the sum of the two given numbers is 32. What is the positive difference between the cubes of the two given numbers? (CGL 2023 Pre)

- (a) 6272 (b) 5832
(c) 4662 (d) 7904

382. During a division, Pranjal mistakenly took as the dividend a number that was 10% more than the original dividend. He also mistakenly took as the divisor a number that was 25% more than the original divisor. If the correct quotient of the original division problem was 25 and the remainder was 0, what was the quotient that Pranjal obtained, assuming his calculations had no error? (CGL 2023 Pre)

- (a) 21.75 (b) 21.25
(c) 28.75 (d) 22

383. Part One While solving a problem, Suhas by mistake took a number as the dividend which was 10% less than the original dividend. He also mistakenly took a number as the denominator which was 20% less than the original denominator. If the correct quotient of the original question of division was 24 and the remainder was 0, then assuming that there was no error in his calculation, what quotient did Suhas get? (CGL 2023 Pre)

- (a) 27 (b) 21.6
(c) 26.4 (d) 30

384. Find the quotient, when the mean proportional of $7\frac{1}{5}$ and 245 is divided by an even prime number? (CGL 2024 Pre)

- (a) 49 (b) 42
(c) 21 (d) 36

SSC CPO

385. The sum of all possible three digit numbers formed by digits 3, 0 and 7 using each digit only once is: (CPO 2018 Pre)

- (a) 2010 (b) 2220
(c) 1990 (d) 2110

386. What is the difference between the greatest four digit and the smallest four digit number using the digits 2, 9, 6 and 5 (each digit can be used only once)? (CPO 2018 Pre)

- (a) 7083 (b) 6993
(c) 6606 (d) 7056

387. The face value of the digit 6 in 16008 is: (CPO 2018 Pre)

- (a) 6 (b) 600
(c) 6000 (d) 60

388. The students of a class donated ₹3,481 towards relief fund. Each student donated an amount equal to the number of students in the class. The number of students in the class is: (CPO 2018 Pre)

- (a) 49 (b) 59
(c) 61 (d) 51

389. A gardener planted 1936 saplings in a garden such that there were as many rows of saplings as the columns. The number of rows planted is: (CPO 2018 Pre)

- (a) 46 (b) 44
(c) 48 (d) 42

390. The students of a class donated a sum of 2,809 to the Fund. Each student donated as many rupees as the number of students in the class. The number of students in the class is: (CPO 2020 Pre)

- (a) 53 (b) 47
(c) 51 (d) 49

391. If the number 647592 is divisible by 88 and if the digits are rearranged in increasing order, then the new number thus formed will be divisible by: (CPO 2022 Pre)

- (a) 22 (b) 66
(c) 44 (d) 03

392. The number 150328 is divisible by 23. If the digits are rearranged in descending order and five times of 13 is subtracted from the new number thus formed, then the resultant number will be divisible by: (CPO 2022 Pre)

- (a) 3 (b) 5
(c) 11 (d) 2

393. The sum of two numbers is 28 and the sum of their squares is 528. Find the square root of the product of both the numbers. (CPO 2024 Pre)

- (a) $8\sqrt{2}$ (b) $2\sqrt{2}$
(c) $4\sqrt{2}$ (d) $6\sqrt{2}$

Number System

394. Arrange the following ratios in ascending order. (CPO 2023 Pre)
3 : 4, 3 : 5, 2 : 11, 7 : 2

- (a) $3 : 5 < 2 : 11 < 3 : 4 < 7 : 2$
(b) $7 : 2 < 3 : 4 < 3 : 5 < 2 : 11$
(c) $2 : 11 < 3 : 5 < 3 : 4 < 7 : 2$
(d) $3 : 4 < 3 : 5 < 2 : 11 < 7 : 2$

395. What must be added to $(x^3 - 3x^2 + 4x - 75)$ to obtain a polynomial that is exactly divisible by $(x - 5)$? (CPO 2023 Pre)

- (a) 5 (b) 6
(c) 4 (d) 2

SSC CHSL

396. Which among the following is an irrational number?

- (1) $\tan 30^\circ \tan 60^\circ$
(2) $\sin 30^\circ$
(3) $\tan 45^\circ$
(4) $\cos 30^\circ$

- (a) (2) (b) (1)
(c) (3) (d) (4)

397. The ratio of the square of a number to the reciprocal of its

cube is $\frac{243}{16807}$, then What is the number? (CHSL 2018 Pre)

- (a) $\frac{2}{7}$ (b) $\frac{7}{3}$
(c) $\frac{3}{7}$ (d) $\frac{7}{7}$

398. For a number, greater than one, the difference between itself and its reciprocal is 20% of the sum of itself and its reciprocal. By how much percentage (nearest to an integer) is the square of the number less than its cube? (CHSL 2018 Pre)

- (a) 122 (b) 18
(c) 81 (d) 33

399. The sum of 17 consecutive numbers is 289. The sum of another 10 consecutive numbers will be whose first term is 5 more than the average of the first set of consecutive numbers, is: (CHSL 2019 Pre)

- (a) 315 (b) 285
(c) 265 (d) 300

400. Find the positive number which when decreased by 20 is equal to 2925 times the reciprocal of the number. (CHSL 2019 Pre)

- (a) 55 (b) 35
(c) 45 (d) 65

401. The difference between two numbers is 3 and the difference between their cubes is 999. Find the difference between their squares. (CHSL 2019 Pre)

- (a) 81 (b) 63
(c) 36 (d) 18

402. Given that $2^{30} + 1$ is completely divisible by a whole number. Which of the following is completely divisible by the same number? (CHSL 2019 Pre)

- (a) $2^{15} + 1$ (b) 5×2^{30}
(c) $2^{30} + 1$ (d) $2^{30} + 1$

403. $2x - 2y$ is a factor of: (CHSL 2019 Pre)

- (a) $4x^2 + 2x - 3y + 9y^2 - 12xy$
(b) $8x^2 + 27y^3$
(c) $4x^2 + 9y^2 + 12xy$
(d) $4x^2 + 2x - 3y + 36y^2 + 12xy$

404. $(ax + by)$ is factor of: (CHSL 2019 Pre)

- (a) $a^2x^2 + 2ab - b^2y^2$
(b) $a^2x^2 + 2abxy + b^2y^2$
(c) $a^2x^2 + 2abx - b^2y^2x$
(d) $a^2x^2 + 2ab - b^2y^2$

405. If a positive integer 'n' is divisible by 3, 5, and 7, then what is the next larger integer divisible by all these numbers? (CHSL 2019 Pre)

- (a) $n + 21$ (b) $n + 35$
(c) $n + 105$ (d) $n + 110$

406. Which of the following numbers will completely divide $7^{81} + 7^{82} + 7^{83}$? (CHSL 2019 Pre)

- (a) 399 (b) 389
(c) 387 (d) 397

407. A man buys 2 apples and 3 kiwi fruits for ₹37. If he buys 4 apples and 5 kiwi fruits for ₹67, then what will be the total cost of 1 apple and 1 kiwi fruit? (CHSL 2019 Pre)

- (a) ₹20 (b) ₹28
(c) ₹18 (d) ₹15

408. The difference between a number and one-third of that number is 228. What is 20% of that number? (CHSL 2019 Pre)

- (a) 68.4 (b) 58.9
(c) 61.8 (d) 72.5

409. The sum of two numbers is 59 and their product is 840. Find the sum of their squares. (CHSL 2019 Pre)

- (a) 2961 (b) 1754
(c) 1801 (d) 1875

410. If $x^3 - 6x^2 + ax + b$ is divisible by $(x^2 - 3x + 2)$, then the values of a and b are: (CHSL 2019 Pre)

- (a) $a = -11$ and $b = 6$
(b) $a = -6$ and $b = -11$
(c) $a = 11$ and $b = -6$
(d) $a = 6$ and $b = 11$

411. By adding 3 and 5 in numerator and denominator of a fraction it becomes $\frac{2}{3}$. If 1 and 3 are subtracted and added from numerator and denominator respectively it becomes $\frac{2}{5}$. Find the fraction. (CHSL 2019 Pre)

- (a) $\frac{5}{7}$ (b) $\frac{6}{7}$
(c) $\frac{7}{6}$ (d) $\frac{7}{5}$

412. Two positive numbers are in the ratio 8:13. If the sum of their squares is 2097, then the sum of the two numbers is: (CHSL 2020 Pre)

- (a) 65 (b) 63
(c) 64 (d) 60

413. A, B and C are three positive numbers such that the average of $\frac{3}{5}$ of A and 30% of B is 13.5, and the average of $\frac{3}{8}$ times of B and 25% of C is

15. If A is equal to $\frac{5}{12}$ times of

C, then the sum of all three numbers A, B and C is: (CHSL 2020 Pre)

- (a) 125 (b) 135
(c) 145 (d) 120

414. What is the product of the largest and the smallest possible values of m for which a number 5 m 83 m 4 ml is divisible by 9? (CHSL 2020 Pre)

- (a) 80 (b) 10
(c) 40 (d) 16

415. If the difference between two numbers is 5 and the difference between their cubes is 1850, then the difference between their squares is: (CHSL 2020 Pre)

- (a) $5\sqrt{482}$ (b) $5\sqrt{483}$
(c) $5\sqrt{484}$ (d) $5\sqrt{485}$

Number System

416. A number is mistakenly multiplied by $\frac{7}{5}$ instead of being multiplied by $\frac{3}{2}$. What is the percentage change in the result due to this mistake? (CHSL 2020 Pre)

- (a) $6\frac{2}{3}\%$ (b) $7\frac{2}{3}\%$
(c) $3\frac{2}{3}\%$ (d) $5\frac{2}{3}\%$

417. Find the greatest among the following options. (CHSL 2021 Pre)

- (a) 0.62 (b) $\frac{2}{3}$
(c) 0.57 (d) $\frac{4}{5}$

418. When a certain number is multiplied by 11, the product is a six-digit number containing only 6s. Find the number that is multiplied by 11. (CHSL 2021 Pre)

- (a) 79365 (b) 78365
(c) 60606 (d) 61661

419. The sum of the first 78 natural numbers from 1 to 78 is divisible by (CHSL 2021 Pre)

- (a) 79 (b) 61
(c) 29 (d) 30

420. The product of three consecutive odd numbers is 693. What is the smallest of the three numbers? (CHSL 2021 Pre)

- (a) 9 (b) 11
(c) 5 (d) 7

421. If $px^3 + x^2 + 3x + q$ is exactly divisible by $(x + 2)$ and $(x - 2)$, then the values of p and q are: (CHSL 2021 Pre)

- (a) $p = -\frac{3}{4}$ and $q = 4$

- (b) $p = \frac{3}{4}$ and $q = 4$

- (c) $p = \frac{3}{4}$ and $q = -4$

- (d) $p = -\frac{3}{4}$ and $q = -4$

422. What is the sum of all two digit odd numbers? (CHSL 2022 Pre)

- (a) 2375 (b) 2475
(c) 2325 (d) 2425

423. If the sum of a number and its reciprocal is 4, find the sum of their squares. (CHSL 2022 Pre)

- (a) 12 (b) 16
(c) 14 (d) 18

424. If sum of squares of two real numbers is 12 and the product of the numbers is 4, find the difference between the numbers. (CHSL 2022 Pre)

(a) 4 (b) 8
(c) 1 (d) 2

425. Which of the following statement is true?

(CHSL 2022 Pre)

I. $\frac{5}{11} > \frac{5}{6}$ II. $\frac{5}{9} > \frac{8}{9}$

III. $\frac{5}{6} > \frac{4}{5}$

(a) Only III (b) Both I and II
(c) Only II (d) Only I

426. The sum of three consecutive even numbers is 126. What is the product of the smallest and the largest number?

(CHSL 2022 Pre)

(a) 1840 (b) 1950
(c) 1760 (d) 1620

427. Which of the following number is the smallest? (CHSL 2022 Pre)

$(14)^{\frac{1}{2}}, (12)^{\frac{1}{3}}, (16)^{\frac{1}{4}}$ & $(25)^{\frac{1}{5}}$

(a) $(14)^{\frac{1}{2}}$ (b) $(25)^{\frac{1}{5}}$

(c) $(16)^{\frac{1}{4}}$ (d) $(12)^{\frac{1}{3}}$

428. What is the sum of the largest and the smallest of the given fractions? (CHSL 2022 Pre)

$\frac{6}{5}, \frac{11}{15}, \frac{4}{7}, \frac{5}{13}$

(a) $\frac{97}{65}$ (b) $\frac{61}{65}$

(c) $\frac{94}{65}$ (d) $\frac{103}{65}$

429. 10 consecutive numbers are given. If the average of the two numbers given in the middle is 13.5, then what is the sum of the first 6 numbers?

(CHSL 2022 Pre)

(a) 58 (b) 59
(c) 67 (d) 69

430. If the sum of three consecutive composite numbers is 36, then what is the product of the three numbers? (CHSL 2022 Pre)

(a) 1460 (b) 1750
(c) 1680 (d) 1820

431. What is the sum of all two digit even numbers? (CHSL 2022 Pre)

(a) 2520 (b) 2470
(c) 2430 (d) 2410

432. Price of 15 chairs is 20 percent more than the price of 8 tables. If price of 2 tables is ₹2400, then what is the price of 15 chairs? (CHSL 2022 Pre)

(a) ₹10,320 (b) ₹12,380
(c) ₹11,520 (d) ₹10,780

433. What is the number of digits required for numbering a book with 428 pages?

(CHSL 2024 Pre)

(a) 1176 (b) 1500
(c) 2000 (d) 988

434. K being any odd number greater than 1, $k^{23} - k$ is always divisible by:

(CHSL 2024 Pre)

(a) 24 (b) 13
(c) 5 (d) 15

435. Out of two number, the first number is three-fourth of the second number. If the average of the reciprocals of the two

numbers is $\frac{7}{72}$, then the sum

of the two number is:

(MTS 2020 Pre)

(a) 21 (b) 22
(c) 25 (d) 20

436. The difference between a positive number and its reciprocal increases by a factor

of $\frac{175}{144}$ when the number is

made to increase by 20%. What is the number? (MTS 2020 Pre)

(a) 7.5 (b) 6
(c) 2.5 (d) 5

437. Let x and y be two positive numbers such that $x > y$. The average of 50% of x and 30% of y is 34.5. If the difference between x and y is 10, then the average of x and y is:

(MTS 2020 Pre)

(a) 85 (b) 80
(c) 75 (d) 64

438. The ratio of the sum, difference and product of two numbers is 11 : 1 : 120. What is the sum of their squares?

(SP Phase-8)

(a) 549 (b) 656
(c) 808 (d) 976

439. In 46.5 weeks, Ram raised ₹2,170 for science research. How much money will he raise in 30 weeks?

SP 2023 Phase-11 Graduation L.

(a) ₹1,400 (b) ₹1,500
(c) ₹1,700 (d) ₹1,600

440. If $2^x = 3^y = 6^z$, then $\frac{1}{x} + \frac{1}{y} - \frac{1}{z}$ is

equal to:

SP 2022 Phase-10 Graduation L.

(a) 1 (b) $-\frac{1}{2}$

(c) 0 (d) $\frac{3}{2}$

441. The sum of three prime numbers is 90. If one of them exceeds another by 30, then one of the numbers is:

SP 2022 Phase-10 Graduation L.

(a) 41 (b) 67
(c) 47 (d) 59

SSC MAINS

442. One of the factors of $(8^{2k} + 5^{2k})$, where k is an odd number, is:

(CGL MAINS 2018)

(a) 84 (b) 86
(c) 89 (d) 88

443. If $(\sqrt{2} + \sqrt{5} - \sqrt{3}) \times k = -12$, then what will be the value of k?

(CGL MAINS 2018)

(a) $(\sqrt{2} + \sqrt{5} - \sqrt{3})(2 + \sqrt{5})$

(b) $(\sqrt{2} + \sqrt{5} + \sqrt{3})(2 - \sqrt{10})$

(c) $(\sqrt{2} + \sqrt{5} + \sqrt{3})(2 - \sqrt{5})$

(d) $(\sqrt{2} + \sqrt{5} + \sqrt{3})$

444. Let a, b and c be the fractions such that $a < b < c$. If c is divided by a, the result is $\frac{5}{2}$, which

exceeds b by $\frac{7}{4}$. If $a + b + c = 1\frac{11}{12}$

then $(c - a)$ will be equal to:

(CGL MAINS 2018)

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$

(c) $\frac{2}{3}$ (d) $\frac{1}{6}$

445. In a school $\frac{4}{9}$ of the number of students are girls and the rest

are boys $\frac{3}{5}$ of the number of

boys are below 12 years of age

and $\frac{5}{12}$ of the number of girls

are 12 years or above 12 years

of age. If the number of

students below 12 years of age

is 480, then $\frac{5}{18}$ of the total

number of students in the

school will be equal to:

(CGL MAINS 2018)

(a) 225 (b) 315
(c) 270 (d) 240

446. Three fraction x, y and z are such that $x > y > z$. When the smallest of them is divided by

the greatest, the result is $\frac{9}{16}$,

which exceeds y by 0.0625. If x

+ y + z = $1\frac{13}{24}$, then the value of

x + z is:

(CGL MAINS 2018)

(a) $\frac{7}{8}$ (b) 1

(c) $\frac{25}{24}$ (d) $\frac{7}{6}$

447. a, b and c are three fraction such that $a < b < c$. If c is divided by a, the result is $\frac{9}{2}$, which exceeds b by $\frac{23}{6}$. The sum of a, b and c is $19\frac{1}{12}$. What is the value of $(2a + b - c)$?

(CGL MAINS 2018)

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$

(c) $\frac{1}{12}$ (d) $\frac{1}{4}$

448. The numerator of a fraction is 3 more than the denominator. When 5 is added to the numerator and 2 is subtracted from the denominator, the

fraction becomes $\frac{8}{3}$. When the

original fraction is divided by

$\frac{1}{2}$, the fraction so obtained is:

(CGL MAINS 2019)

(a) $\frac{2}{3}$ (b) $\frac{1}{4}$

(c) $\frac{1}{2}$ (d) $\frac{3}{4}$

449. Let, $x = \left(\frac{\sqrt{1875}}{\sqrt{3888}} + \frac{\sqrt{1200}}{\sqrt{768}} \right) \times$

$\frac{\sqrt{175}}{\sqrt{1792}}$. Then $\sqrt{x}$ is equal to:

(CGL MAINS 2019)

(a) $\frac{7}{12}$ (b) $\frac{4}{9}$

(c) $\frac{5}{12}$ (d) $\frac{5}{9}$

450. When 1062, 1134 and 1182 are divided by the greatest number x, the remainder in each case is y. What is the value of $(x - y)$?

(CGL MAINS 2019)

(a) 19 (b) 17
(c) 16 (d) 18

451. If $\frac{1}{4.263} = 0.2346$, find the

value of $\frac{1}{0.0004263}$.

(CGL MAINS 2019)

(a) 4.263 (b) 2346
(c) 4263 (d) 2.346

452. The sum of three numbers is 280. If the ratio between the first and second numbers is 2 : 3 and the ratio between second and third numbers is 4 : 5, then find the second number.

(CGL MAINS 2019)

(a) 96 (b) 90
(c) 86 (d) 80

453. If $\frac{b}{a} = 0.7$, find the value of $\frac{a - b + 11}{a + b + 34}$.

(CGL MAINS 2019)

(a) 1 (b) 0.5
(c) 0.3 (d) 0.2

454. If $847 \times 385 \times 675 \times 3025 = 3^a \times 5^b \times 7^c \times 11^d$, then the value of $ab - cd$ is:

(CGL MAINS 2019)

(a) 4 (b) 5
(c) 1 (d) 7

455. Three fractions x, y and z are such that $x > y > z$. When the smallest of them is divided by

the greatest, the result is $\frac{9}{16}$

which exceeds y by 0.0625. If x

+ y + z = $2\frac{2}{12}$, then what is the

value of x + z?

(CGL MAINS 2019)

(a) $\frac{7}{4}$ (b) $\frac{3}{4}$
(c) $\frac{5}{4}$ (d) $\frac{1}{4}$

456. Let, x, y, z be fractions such that $x < y < z$. If z is divided by x, the

result is $\frac{5}{2}$, which exceeds y

by $\frac{7}{4}$. If $x + y + z = 1\frac{11}{12}$, then the

ratio of $(z - x) : (y - x)$ is:

(CGL MAINS 2019)

(a) 6 : 5 (b) 9 : 5
(c) 5 : 6 (d) 5 : 9

457. The sum of three fractions A, B and C is $\frac{121}{60}$. Where is $A > B > C$. When C is divided by B, the

resulting fraction is $\frac{9}{10}$ to which

exceeds A by $\frac{3}{20}$. What is the

difference between B and C?

(CGL MAINS 2019)

(a) $\frac{1}{15}$ (b) $\frac{1}{10}$

(c) $\frac{3}{10}$ (d) $\frac{7}{15}$

458. If $\sqrt[3]{N}$ lies between 6 and 7, where N is an integer then how many values N can take?

(CGL MAINS 2021)

(a) 126 (b) 127
(c) 128 (d) 125

459. x, y and z are distinct prime numbers where $x < y < z$. If $x + y + z = 70$, then what is the value of z?

(CGL MAINS 2021)

(a) 29 (b) 43
(c) 31 (d) 37

460. Which fraction among the following is the least?

(CGL MAINS 2021)

$\frac{5}{11}, \frac{7}{12}, \frac{8}{13}, \frac{9}{17}$

(a) $\frac{5}{11}$ (b) $\frac{7}{12}$

(c) $\frac{9}{17}$ (d) $\frac{8}{13}$

461. Find the average of the cubes of the first five natural numbers.

(CGL MAINS 2022)

(a) 35 (b) 40
(c) 45 (d) 50

462. The average of the square of first five even natural numbers is:

(CHSL 2022 Mains)

(a) 40 (b) 44
(c) 42 (d) 46

463. What is the rightmost digit preceding the zeros in the value of 30^{100} ? (UPSC CSAT 2024)
(a) 1 (b) 3
(c) 7 (d) 9
464. A can contains 399X litres of petrol and a can contains 532Y litres of diesel. They are to be bottled in bottles of equal size so that whole of petrol and diesel would be separately bottled. The bottle capacity in terms of litres is an integer. How many different bottle sizes are possible? (UPSC CSAT 2024)
(a) 3 (b) 4
(c) 5 (d) 6
465. On January 1st 2023, a person saved ₹1. On January 2nd 2023, he saved ₹2 more than that on the previous day. On January 3rd, 2023, he saved Rs 2 more than that on the previous day and so on. At the end of which date was his total savings a perfect square, as well as a perfect cube?
(UPSC CSAT 2024)
(a) 7th January (b) 8th January
(c) 9th January (d) Not possible
466. Consider the following statements in respect of the sum $S = x + y + z$, where x , y and z are distinct prime numbers each less than 10?
I. The unit digit of S can be 0.
II. The unit digit of S can be 9.
III. The unit digit of S can be 5.
Which of the following statements given above are correct? (UPSC CSAT 2024)
(a) I and II only
(b) II and III only
(c) I and III only
(d) I, II and III
467. How many consecutive zeros are there at the end of the integer obtained in the product $1^2 \times 2^4 \times 3^6 \times 4^8 \times \dots \times 25^{50}$?
(UPSC CSAT 2024)
(a) 50 (b) 55
(c) 100 (d) 200
468. Three numbers x , y , z are selected from the set of the first seven natural numbers such that $x > 2y > 3z$. how many such distinct triplets (x, y, z) are possible? (UPSC CSAT 2024)
(a) One triplet
(b) Two triplet
(c) Three triplet
(d) Four triplet
469. What is the number of five used in numbering a 260-page book? (UPSC CSAT 2024)
(a) 55 (b) 57
(c) 56 (d) 60
470. Let p and q be positive integers satisfying $p < q$ and $p + q = k$. what is the smallest value of k that does not determine p and q uniquely? (UPSC CSAT 2024)
(a) 3 (b) 4
(c) 5 (d) 6
471. A question is given followed by two statements I and II. Consider the Question and the Statements.
Question:
What are the unique values of x and y , where x and y are distinct natural numbers?
Statement-I: $\frac{x}{y}$ is odd
Statement-II: $xy = 12$
Which one of the following is correct in respect to the above Question and the Statements?
(UPSC CSAT 2024)
(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone.
(b) The Question can be answered by using either Statement alone.
(c) The question can be answered by using both the Statements together, but cannot be answered using either Statement alone.
(d) The Question cannot be answered even by using both the Statements together.
472. Let p , q , r and s be distinct positive integers.
Let p , q be odd and r , s be even consider the following statements:
I. $(p - r)^2 (qs)$ is even.
II. $(q - s)q^2s$ is even.
III. $(q + r)^2 (p + s)$ is odd
Which of the statement given above are correct?
(UPSC CSAT 2024)
(a) I and II only
(b) II and III only
(c) I and III only
(d) I, II, III

473. The least common multiple of two prime numbers x and y ($x < y$) is 141. Then what is the value of $(y - 5x)$?
(SSC GD 2024)
(a) 30 (b) 32
(c) 40 (d) 38

ANSWER KEY - NUMBER SYSTEM

1. (b) 2. (b) 3. (c) 4. (d) 5. (b) 6. (d) 7. (b) 8. (a) 9. (c) 10. (d)
11. (a) 12. (a) 13. (a) 14. (c) 15. (d) 16. (c) 17. (c) 18. (b) 19. (c) 20. (d)
21. (b) 22. (c) 23. (b) 24. (d) 25. (a) 26. (b) 27. (c) 28. (b) 29. (b) 30. (d)
31. (b) 32. (d) 33. (a) 34. (b) 35. (b) 36. (b) 37. (d) 38. (c) 39. (c) 40. (c)
41. (b) 42. (d) 43. (d) 44. (a) 45. (a) 46. (b) 47. (a) 48. (b) 49. (c) 50. (b)
51. (b) 52. (b) 53. (c) 54. (a) 55. (a) 56. (c) 57. (b) 58. (c) 59. (a) 60. (a)
61. (c) 62. (d) 63. (a) 64. (c) 65. (a) 66. (c) 67. (b) 68. (d) 69. (d) 70. (b)
71. (b) 72. (c) 73. (d) 74. (c) 75. (b) 76. (b) 77. (a) 78. (b) 79. (b) 80. (a)
81. (b) 82. (c) 83. (d) 84. (a) 85. (c) 86. (a) 87. (c) 88. (c) 89. (d) 90. (c)
91. (d) 92. (d) 93. (d) 94. (a) 95. (b) 96. (a) 97. (d) 98. (c) 99. (b) 100. (d)
101. (a) 102. (b) 103. (b) 104. (a) 105. (c) 106. (b) 107. (b) 108. (b) 109. (c) 110. (a)
111. (b) 112. (c) 113. (c) 114. (a) 115. (c) 116. (d) 117. (b) 118. (d) 119. (d) 120. (c)
121. (a) 122. (b) 123. (a) 124. (a) 125. (d) 126. (d) 127. (d) 128. (c) 129. (c) 130. (a)
131. (a) 132. (c) 133. (a) 134. (b) 135. (d) 136. (a) 137. (d) 138. (a) 139. (d) 140. (a)
141. (b) 142. (b) 143. (d) 144. (c) 145. (b) 146. (a) 147. (d) 148. (b) 149. (d) 150. (a)
151. (b) 152. (d) 153. (a) 154. (a) 155. (c) 156. (a) 157. (c) 158. (d) 159. (c) 160. (b)
161. (b) 162. (b) 163. (b) 164. (d) 165. (b) 166. (a) 167. (d) 168. (c) 169. (a) 170. (d)
171. (a) 172. (c) 173. (d) 174. (b) 175. (c) 176. (a) 177. (c) 178. (c) 179. (b) 180. (a)
181. (b) 182. (a) 183. (d) 184. (a) 185. (d) 186. (a) 187. (c) 188. (b) 189. (d) 190. (a)
191. (a) 192. (c) 193. (a) 194. (c) 195. (c) 196. (c) 197. (b) 198. (c) 199. (d) 200. (b)
201. (d) 202. (b) 203. (c) 204. (a) 205. (c) 206. (b) 207. (c) 208. (c) 209. (a) 210. (b)
211. (a) 212. (a) 213. (a) 214. (d) 215. (b) 216. (b) 217. (c) 218. (b) 219. (b) 220. (b)
221. (d) 222. (b) 223. (d) 224. (c) 225. (a) 226. (d) 227. (a) 228. (c) 229. (a) 230. (c)
231. (c) 232. (a) 233. (b) 234. (a) 235. (c) 236. (d) 237. (a) 238. (d) 239. (d) 240. (d)
241. (d) 242. (a) 243. (d) 244. (b) 245. (d) 246. (c) 247. (a) 248. (b) 249. (d) 250. (b)
251. (a) 252. (d) 253. (b) 254. (d) 255. (a) 256. (b) 257. (d) 258. (c) 259. (d) 260. (a)
261. (b) 262. (d) 263. (d) 264. (d) 265. (b) 266. (c) 267. (b) 268. (b) 269. (a) 270. (c)
271. (b) 272. (c) 273. (d) 274. (d) 275. (b) 276. (a) 277. (d) 278. (c) 279. (a) 280. (b)
281. (c) 282. (d) 283. (d) 284. (c) 285. (d) 286. (a) 287. (b) 288. (c) 289. (d) 290. (b)
291. (c) 292. (a) 293. (c) 294. (d) 295. (b) 296. (c) 297. (c) 298. (b) 299. (d) 300. (d)
301. (c) 302. (d) 303. (d) 304. (d) 305. (c) 306. (c) 307. (c) 308. (b) 309. (d) 310. (d)
311. (d) 312. (a) 313. (c) 314. (d) 315. (d) 316. (b) 317. (b) 318. (c) 319. (d) 320. (a)
321. (c) 322. (b) 323. (b) 324. (a) 325. (d) 326. (b) 327. (b) 328. (a) 329. (d) 330. (b)
331. (c) 332. (a) 333. (d) 334. (c) 335. (b) 336. (c) 337. (a) 338. (a) 339. (d) 340. (a)
341. (b) 342. (c) 343. (d) 344. (c) 345. (a) 346. (d) 347. (c) 348. (a) 349. (c) 350. (c)
351. (c) 352. (b) 353. (b) 354. (b) 355. (d) 356. (d) 357. (a) 358. (d) 359. (d) 360. (a)
361. (a) 362. (a) 363. (a) 364. (c) 365. (a) 366. (b) 367. (a) 368. (a) 369. (b) 370. (c)
371. (b) 372. (d) 373. (c) 374. (c) 375. (b) 376. (d) 377. (c) 378. (b) 379. (d) 380. (a)
381. (a) 382. (d) 383. (a) 384. (c) 385. (d) 386. (a) 387. (a) 388. (b) 389. (b) 390. (a)
391. (d) 392. (b) 393. (a) 394. (c) 395. (a) 396. (d) 397. (c) 398. (b) 399. (c) 400. (d)
401. (b) 402. (d) 403. (a) 404. (b) 405. (c) 406. (a) 407. (d) 408. (a) 409. (c) 410. (c)
411. (a) 412. (b) 413. (a) 414. (d) 415. (d) 416. (a) 417. (d) 418. (c) 419. (a) 420. (d)
421. (d) 422. (b) 423. (c) 424. (d) 425. (a) 426. (c) 427. (b) 428. (d) 429. (d) 430. (c)
431. (c) 432. (c) 433. (a) 434. (a) 435. (a) 436. (d) 437. (a) 438. (d) 439. (a) 440. (c)
441. (d) 442. (c) 443. (b) 444. (b) 445. (a) 446. (c) 447. (d) 448. (b) 449. (c) 450. (d)
451. (b) 452. (a) 453. (b) 454. (b) 455. (a) 456. (a) 457. (a) 458. (a) 459. (d) 460. (a)
461. (c) 462. (b) 463. (d) 464. (b) 465. (b) 466. (c) 467. (d) 468. (d) 469. (c) 470. (c)
471. (c) 472. (d) 473. (b)

Smart Tricks & Solutions

Divisibility Rule

SSC CGL

1. (b) $\frac{985x3678y}{72}$
 $72 = 8 \times 9$
 so we apply divisibility rule of 8 & 9
 Divisibility rule for 8 $\rightarrow$ last 3 digits must be divisible by 8.
 $78y + 8 \Rightarrow$ here y must be 4
 Divisibility rule for 9 $\rightarrow$ sum of digits must be divisible by 9.
 $9+8+5+x+3+6+7+8+4$
 $50+x$
 here x must be 4
 $x=4, y=4$
 $\Rightarrow 4x-3y = 16-12=4$
2. (b) $\frac{2094x843y2}{88}$
 88 is multiple of 11 & 8 so we check divisibility rule for 11 and 8
 Divisibility rule for 8 $\rightarrow$ last 3 digits must be divisible by 8.
 $3y2 + 8 \Rightarrow$ here y must be 5
 Divisibility rule for 11 $\rightarrow$ The difference between the sum of the digits in the odd places and the sum of the digits in the even places must be zero or multiple of 11.
 $2+9+x+4+5$ $0+4+8+3+2$
 $20+x$ 17
 here x must be 8
 $x = 8$ (largest)
 $y = 5$
 $\Rightarrow 5x - 7y = 5 \times 8 - 7 \times 5 = 5$
3. (c) $897359y7x2$
 For a number to be divisible by 72, that number should be divisible by 9 as well as 8.
 Divisibility rule for 8 $\rightarrow$ last 3 digits must be divisible by 8.
 $\frac{7 \times 2}{8} \Rightarrow x = 5$ (for max value of y)
 Divisibility rule for 9 $\rightarrow$ sum of digits must be divisible by 9.
 $897359y752 \Rightarrow \frac{55+y}{9}$
 y must be 8
 $\Rightarrow 3x - y = 3(5) - 8 = 7$

4. (d) $5y5884805x6$
 $72 = 8 \times 9$
 Divisibility rule for 8 $\rightarrow$ last 3 digits must be divisible by 8.
 $\frac{5x6}{8} \Rightarrow x = 3$ or 7
 Divisibility rule for 9 $\rightarrow$ sum of digits must be divisible by 9.
 $5+y+5+8+8+4+8+0+5+3+6$
 $52+y$
 If we take $x = 3$ then $y = 2$
 $\sqrt{xy} = \sqrt{6}$
 If we take $x = 7$ then $y = 7$ but in que. $x \neq y$
5. (b) $15x1y2$
 $44 = 4 \times 11$
 Divisibility rule for 4 $\rightarrow$ last 2 digits must be divisible by 4.
 $\frac{y2}{4} \Rightarrow$ here y must be 1
 Divisibility rule for 11 $\rightarrow$ The difference between the sum of the digits in the odd places and the sum of the digits in the even places must be zero or multiple of 11.
 $\frac{15x1y2}{5+1+2}$ $\frac{1+x+1}{2+x}$
 8 $2+x$
 $x=6$
 $\Rightarrow x+y = 6+1=7$
6. (d) $\frac{517x324}{12}$
 12 is multiple of 3×4 so we will check 3 and 4 divisibility rule
 Divisibility rule for 4 $\rightarrow$ last 2 digits must be divisible by 4.
 24 is completely divisible by 4.
 Divisibility rule for 3 $\rightarrow$ sum of digits must be divisible by 3.
 $5+1+7+x+3+2+4 = \text{multiple of } 3$
 $\frac{22+x}{3} \Rightarrow x = 2$
7. (b) Remainder is 12 it means divisor must be greater than 12.
 $\frac{720+12}{x}$
 720 , must be divisible by x , or x is a factor of 720.
 $720 = 2^4 \times 3^2 \times 5^1$
 Total factors of 720
 $= (4+1) \times (2+1) \times (1+1) = 30$
 Factors till 12
 $= 1, 2, 3, 4, 5, 6, 8, 9, 10, 12 = 10$
 $x = 30 - 10 = 20$

8. (a) $x35624$ is divisible by 11.
 Apply Rule of divisibility of 11.
 $x+5+2=3+6+4$
 $x+7=13$
 $x=6$
 $1257y4$ is divisible by 72.
 $72 = 9 \times 8$
 A number must be divisible by 9 when the sum of digits divisible by 9.
 For 8 - Last 3 digits must be divisible by 8.
 $1+2+5+7+y+4$
 $19+y$ (sum must be 27)
 $\therefore y=8$
 y must be 8
 $\Rightarrow 5x-2y$
 $= 5 \times 6 - 2 \times 8$
 $= 30 - 16 = 14$
9. (c) $99 = 11 \times 9$
 A number is divisible by 99 when that number is separately divisible by 11 & 9.
 Divisibility Rule of 11 $\rightarrow$ The difference of sum of even place digits and sum of odd place digits either zero or a multiple of 11.
 Divisibility Rule of 9 $\rightarrow$ sum of a number must be divisible by 9.
 $\frac{708x6y8z9}{7+8+6+8+9 = 0+x+y+z}$
 $38 - x+y+z$
 $\frac{38-(x+y+z)}{11}$
 $\frac{38+x+y+z}{9} = (\text{sum of no.})$
 Now go through option (3)
 $\frac{38-16}{11} = \frac{22}{11}$ (divisible)
 $\frac{38+16}{9} = \frac{54}{9}$ (divisible)
 $\Rightarrow x+y+z = 16$
10. (d) Divisibility rule of 4 $\rightarrow$ last two digits of any no must be divisible by 4.
 $\therefore 2365 * 4$
 Required digits = * 4
 for divisible by 4 possible digits are 0, 2, 4, 6, 8.
 but in question largest digit is asked so ans will be 8.

11. (a) L.C.M of 3, 7, 11 = 231
 Largest possible value of $247xy$ is = 24799
 When we divide 24799 by 231 we obtain 82 as a remainder
 So required number
 $= 24799 - 82 = 24717$
 After comparing
 $x=1$
 $y=7$
 $\Rightarrow 2y-8x = 14-8 = 6$

12. (a) Divisibility rule for 3 $\rightarrow$ sum of digits must be divisible by 3.
 $5306P2$

$$\begin{array}{c} 2 \quad 8 \\ 5+3+0+6+P+2 \\ = \frac{16+P}{3} \end{array}$$

$$P = 2, 5, 8$$

$$\text{Difference} = (8)^2 - (2)^2 = 60$$

13. (a) Divisibility rule for 3 $\rightarrow$ sum of digits must be divisible by 3.

$$\begin{array}{c} 45082 \text{ K} \\ \swarrow \quad \searrow \\ G \quad S \\ 8 \quad 2 \end{array}$$

$$= (G)^2 + (S)^2 = 64+4 = 68$$

14. (c) If the six digits number divisible by 7, 11, 13 then the first three digits and the last three digits are same
 $823p2q$
 $\downarrow \quad \downarrow$
 $8 \quad 3$

$$\Rightarrow p-q = 8-3 = 5$$

15. (d) LCM of (3, 7, 11) = 231
 Let 593ab is the greatest no. is with values of a & $b = 59399$
 On dividing by 231
 Remainder $\Rightarrow 32$
 So, the actual no = 231×257
 $= 59367$

$$a=6, b=7$$

$$\Rightarrow a^2-b^2+ab = 36-49+42 = 29$$

16. (c) $abc = (100a+10b+c)$
 $cab = (100c+10a+b)$
 $bca = (100b+10c+a)$
 Calculate according to question
 $(100a+10b+c) + (100c+10a+b) + (100b+10c+a)$
 $= 111a+111b+111c$
 $= 111(a+b+c)$
 $111 = 3 \times 37$
 The sum of 3 digit numbers abc , cab and bca is not divisible by '31'.

17. (c) Divisibility rule for 11 $\rightarrow$ The difference between the sum of the digits at the odd places and the sum of the digits at the even places must be zero or multiple of 11.

$$\begin{array}{l} 9m2365248 \\ 27-(12+m) = 0 \\ 15-m = m-4 \\ \Rightarrow m^2-n^2 = 16 \times 4 = 64 \end{array}$$

22. (c) $\frac{234a5b}{2 \times 11}$ But not 5

18. (b) $23a68b$
 Divisibility rule for 3 $\rightarrow$ sum of digits must be divisible by 3.
 Divisibility rule for 9 $\rightarrow$ sum of digits must be divisible by 9.
 $\therefore$ Check options
 digit sum of (a) and (c) is divisible by 9.
 Out of (b) and (d) both numbers are divisible by 3 but 239685 is greater.
 $\therefore 239685$

19. (c) Divisible by 7 Make pair of 3 digits from right side.
 Add alternate pairs
 Take difference of the sum if difference is divisible by 7 then the number is divisible by 7, otherwise not divisible by 7.
 $x8942y4$ $56 \rightarrow 7 \times 8$

$$\begin{array}{l} \frac{2y4}{8} \rightarrow y=2, 6 \\ \therefore y \text{ (largest)} = 6 \\ x894264 \\ 894-(264+x) \\ \frac{630-x}{7} \therefore x=7 \\ \therefore x^2+y = 49+6 = 55 \end{array}$$

20. (d) $\frac{7x79251y8}{9 \times 4}$

$$\begin{array}{l} \frac{y8}{4} \Rightarrow y=0, 2, 4, 6, 8 \\ \therefore y=8 \text{ (largest)} \\ \frac{7x7925188}{9} \\ = \frac{47+x}{9} \Rightarrow \frac{2+x}{9} \\ \therefore x=7 \\ \therefore 10x^2-3y^2 = 490-192 = 298 \end{array}$$

21. (b) $88 = 11 \times 8$
 Divisibility rule for 8 $\rightarrow$ last 3 digits must be divisible by 8.

$$\begin{array}{l} \frac{n48}{8} \Rightarrow n=2 \text{ (for smallest)} \\ \text{Divisibility rule for 11} \rightarrow \text{The difference between the sum of the digits in the odd places and the sum of the digits in the} \end{array}$$

even places must be zero or multiple of 11.

$$\begin{array}{l} 9m2365248 \\ 27-(12+m) = 0 \\ 15-m = m-4 \\ \Rightarrow m^2-n^2 = 16 \times 4 = 64 \end{array}$$

22. (c) $\frac{234a5b}{2 \times 11}$ But not 5

$234a5b$
 $11-(3+a+b) = 0$ or div. by 11
 $8-(a+b) = 0$ or div. by 11
 Divisibility rule for 11 $\rightarrow$ The difference between the sum of the digits in the odd places and the sum of the digits in the even places must be zero or multiple of 11.
 $\therefore a+b$ should be 8
 $b \neq 0$ (not divisible by 5)
 $b = 1$ (not divisible by 2)
 $\therefore$ From remaining numbers **234652** is greatest.

23. (b) $6 = 2 \times 3$
 $k \neq 3, 9, k \neq 0, 5$
 $\therefore$ for divisibility by 2 last digit should be 0 or any even no.
 $k = 6$ or 4
 check 4 $\rightarrow$
 digit sum = $15+k$
 $k \neq 4 \therefore$ for divisible by 3 digit sum should be divisible by 3
 $\therefore k = 6$

24. (d) $3 \rightarrow$ Digit sum should be divisible by 3
 $\frac{17+p+q}{3}$
 $\therefore p+q = 4, 7, 10, 13, 16$
 $11 \rightarrow \frac{6+4+p+q}{10}$
 $(6+4+q) - (7+p)$
 $(10+q) - (7+p)$
 $(p-q) = 3$

- Only option (d) has $p-q = 3$
 $\therefore p = 5$ and $q = 2$

25. (a) $3^{30} + 9^{26} + 27^{18} + 9^{28} + 9^{29}$
 $\Rightarrow 9^{25} + 9^{26} + 9^{27} + 9^{28} + 9^{29}$
 $\Rightarrow 9^{25} (1 + 9 + 9^2 + 9^3 + 9^4)$
 $\Rightarrow 9^{25} (1 + 9 + 81 + 729 + 6561)$
 $\Rightarrow 9^{25} \times 7381$ which is divisible by 11.

26. (b) $\frac{42 \times 25 \times 54 \times 135}{3a}$

Out of the given options the maximum value of $a = 7$

$$\therefore \frac{42 \times 25 \times 54 \times 135}{21}$$

Which divides it completely

27. (c) $75 \rightarrow 25 \times 3$

$25 \rightarrow$ Last two digit should be divisible by 25

$$\frac{7q}{25}, q = 5$$

$3 \rightarrow$ digit sum should be divisible by 3

$$\frac{678p375}{3}$$

$$\Rightarrow \frac{36+p}{3}, P = 3$$

$$\therefore p = 3, q = 5$$

28. (b) Any six digit number of the form $abcabc$ is always divisible by 1001.

29. (b) $33 \rightarrow 3 \times 11$

$$11 \rightarrow \frac{52A6B7C}{11}$$

$$(5 + A + B + C) - (2 + 6 + 7)$$

$$(A + B + C) - 10$$

$$\therefore A + B + C = 10, 0$$

$$\frac{52A6B7C}{3}$$

$$\Rightarrow \frac{20 + (A + B + C)}{3}$$

$$\therefore (A + B + C) = 10$$

$\therefore$ Values of (A, B, C) are $(2, 3, 5)$

$\therefore$ Maximum value of

$$2A + 3B + C$$

$$= 2 \times 3 + 3 \times 5 + 2$$

$$= 23$$

30. (d) $6^{25} + 6^{26} + 6^{27} + 6^{28}$

$$= 6^{25} (1 + 6 + 6^2 + 6^3)$$

$$= 6^{25} (1 + 6 + 36 + 216)$$

$$= 6^{25} \times 259$$

The given no. is divisible by

$$259$$

(b) $36 \rightarrow 9 \times 4$

Divisibility by 4 $\rightarrow$ Last 2 digit should be divisible by 4.

Divisibility by 9 $\rightarrow$ digit sum of the number should be divisible by 9.

$29412 \rightarrow$ Last two digit is divisible by 4

Digit sum = $2 + 9 + 4 + 1 + 2 = 18$

18 is divisible by 9

$\therefore 29412$ is divisible by 36

32. (d) Divisibility by 11 $\rightarrow$

Took the sum of the odd places digits and even place digits.

Take their difference if difference is zero or divisible by 11 the number will be divisible by 11.

$$\frac{9592330}{11}$$

$$\text{Odd place sum} = 9 + 9 + 3 + 0 = 21$$

$$\text{Even place sum} = 5 + 2 + 3 = 10$$

$$\text{Difference} = 21 - 10 = 11$$

Which is divisible by 11

$\therefore 9592330$ is divisible by 11.

33. (a) $33 \rightarrow 3 \times 11$

$\therefore$ Number should be divisible by both 3 and 11.

$$35508 \rightarrow \text{digit sum} = 21$$

$\therefore$ Divisible by 3

$$\frac{35508}{11}$$

$$\text{Odd place sum} = 3 + 5 + 8 = 16$$

$$\text{Even place sum} = 5 + 0 = 5$$

$$\text{Difference} = 16 - 5 = 11$$

$\therefore 35508$ is divisible by 33.

34. (b) $99 \rightarrow 9 \times 11$

$$60687 \rightarrow \text{digit sum} = 27$$

$\therefore$ divisible by 9

$$\frac{60687}{11}$$

$$\text{Odd place sum} = 6 + 6 + 7 = 19$$

$$\text{Even place sum} = 0 + 8 = 8$$

$$\text{Difference} = 19 - 8 = 11$$

$\therefore 60687$ is divisible by 99

35. (d) $44 \rightarrow 11 \times 4$

$4 \rightarrow$ Last two digit divisible by 4

$$\frac{54736}{4}, R = 0$$

$$\frac{54736}{11}$$

$$\text{Odd place sum} = 5 + 7 + 6 = 18$$

$$\text{Even place sum} = 4 + 3 = 7$$

$$\text{Difference} = 18 - 7 = 11$$

Which is divisible by 11

$\therefore 54736$ is divisible by 44

36. (b) $\frac{abba}{4}, a < b$

$4 \rightarrow$ Last two digit should be divisible by 4

$$\frac{ba}{4}$$

$$\frac{b}{4} \quad \frac{a}{4}$$

$$9 \quad 2, 6$$

$$8 \quad 4,$$

$$7 \quad 6, 2$$

$$6 \quad 4,$$

$$5 \quad 2$$

$$3 \quad 2$$

$\therefore 8$ numbers possible.

37. (d) Sum of six consecutive numbers is always divisible by 3.

Hence that number will always be divisible by 3

Example:- 123456

$$\text{Sum} = 21$$

$\therefore$ divisible by 3.

38. (c) Digit sum of 5826 $\Rightarrow 21$

$\therefore$ Not divisible by 9

$$\text{Digit sum of } 5964 \Rightarrow 24$$

$\therefore$ Not divisible by 9

$$\text{Digit sum of } 6039 \Rightarrow 18$$

$$\text{Digit sum of } 6336 \Rightarrow 18$$

Digit sum of 6489 $\Rightarrow 27$

Digit sum of 6564 $\Rightarrow 21$

$\therefore$ Not divisible by 9

$$\text{Digit sum of } 6867 \Rightarrow 27$$

Digit sum of 6960 $\Rightarrow 21$

$\therefore$ Not divisible by 9

$\therefore 4$ numbers are not divisible by 9

39. (c) $\frac{45yz0}{8 \times 5}$

Divisibility rule for 8 $\rightarrow$ last 3 digits must be divisible by 8.

$$\frac{yz0}{8} \Rightarrow \frac{880}{8}$$

$$\therefore \text{Maximum value of } (y + z) = 8 + 8 = 16$$

40. (c) $\frac{723 \times 56 \times 6}{2 \times 3}$

Sum of digits should be divisible by 3

$$\frac{29 + * + *}{3}$$

$$\therefore \frac{29 + 2 + 2}{3} = \frac{33}{3} \quad (\checkmark)$$

$\therefore$ Smallest value of $*$ = 2

41. (b) $\frac{321y72}{2 \times 3}$

Sum of digits should be divisible by 3

$$\frac{15 + y}{3}$$

$\therefore$ Least value of $y = 0$

42. (d) $\frac{436P5}{3}$

Sum of digits should be divisible by 3

$$\frac{18 + P}{3}$$

$\therefore$ Largest value of $P = 9$

43. (d) 429714

Sum of digits = $4 + 2 + 9 + 7 + 1 + 4 = 27$

$\therefore$ It is divisible by 3

Also unit digit is 4, hence

It is divisible by 2

$\therefore$ Since it is divisible by 2 and 3 both, hence it is also divisible by 6

44. (a) Sum of first n natural

$$\text{numbers} = \frac{n(n+1)}{2}$$

Sum of first 12 natural

$$\text{numbers} = \frac{12 \times 13}{2} = 6 \times 13$$

$\therefore$ This is divisible by 6

45. (a) $7^{15} + 7^{16} + 7^{17}$

$$\Rightarrow 7^{15} (1 + 7 + 7^2)$$

$$\Rightarrow 7^{15} (1 + 7 + 49)$$

$$\Rightarrow 7^{15} \times 57$$

$$\Rightarrow 7^{15} \times 19 \times 3$$

$\therefore$ This is divisible by 3

46. (b) $\frac{2^{2n} + 2n}{6}$

for $n = 1, 2^n + 2n = 6$

Which is divisible by 6

$\therefore$ Least value of $n = 1$

47. (a) $8^m - 1$

$$\text{put } m = 2$$

$$8^2 - 1 = 63$$

Which is divisible by 63

48. (b) Divisibility rule for 11 $\rightarrow$

The difference between the sum of the digits in the odd places and the sum of the digits in the even places must be zero or multiple of 11.

$$A \Rightarrow \frac{712712}{11}$$

$$(7 + 2 + 1) - (1 + 7 + 2)$$

$$(10) - (10) = 0$$

It is divisible by 11

$$B \Rightarrow \frac{177210}{11}$$

$$(1 + 7 + 7) - (2 + 1 + 0)$$

$$(15) - (3) = 12$$

It is not divisible by 11

$$C \Rightarrow \frac{64614}{11}$$

$$(6 + 6 + 4) - (4 + 1)$$

$$(16) - (5) = 11$$

It is divisible by 11

$$D \Rightarrow \frac{756148}{11}$$

$$(7 + 6 + 4) - (5 + 1 + 8)$$

$$(17) - (14) = 3$$

D is not divisible by 11

49. (c) $\frac{6745K2}{3}$

Sum of digits be divisible by 3

$$\frac{24 + K}{3}$$

$\therefore$ Greatest value of $K = 9$

50. (b) Divisibility rule for 11 $\rightarrow$

The difference between the sum of the digits in the odd places and the sum of the digits in the even places must be zero or multiple of 11.

$$(7 + 7 + 7 + 5) - (2 + 8 + 4)$$

$$(21 + 5) - (14)$$

$$26 - 14 = 12$$

This should be 25 to get diff 11

$\therefore 1$ should be subtracted

51. (b) $\frac{62684 \times \#}{8 \times 5}$

$$\# \rightarrow 0 \text{ or } 5$$

But it is also divisible by 8.

So, 5 is not possible

$$\therefore \# = 0$$

$$\frac{4 \times 0}{8}, * = 4$$

$$\therefore 4 \text{ and } 0$$

52. (b) Divisibility rule for 11 $\rightarrow$

The difference between the sum of the digits in the odd places and the sum of the digits in the even places must be zero or multiple of 11.

$$\frac{7845K854}{11}$$

$$(7 + 4 + K + 5) - (8 + 5 + 8 + 4)$$

$$(16 + K) - (25) = 0$$

$$\therefore K = 9$$

53. (c) $\frac{128041}{11}$

$$(1 + 8 + 4) - (2 + 0 + 1)$$

$$(13) - (3) = 10$$

$\therefore$ It is not divisible by 11

54. (a) $\frac{91876a2}{8}$

Divisibility rule for 8 $\rightarrow$ last 3 digits must be divisible by 8.

$$\frac{6a2}{8}$$

Smallest value of $a = 3$

$$\frac{632}{8}, \text{Remainder} = 0$$

55. (a) $\frac{72864 \times \$}{8 \times 5}$

$\$ \rightarrow$ It should be 0 or 5.

But no is also divisible by 8.

Hence $\$ = 0$

$$\text{Divisible by } 8 \rightarrow \frac{4 \times 0}{8}, * = 4$$

$$\therefore 4 \text{ and } 0$$

56. (c) Divisibility rule for 11 $\rightarrow$

The difference between the sum of the digits in the odd places and the sum of the digits in the even places must be zero or multiple of 11.

$$(5 + 9 + 7) - (8 + 2 + 0)$$

$$(21) - (10) = 11$$

$\therefore 589270$ is divisible by 2, 5, 10 and 11

$$\frac{589270}{11}$$

$$(5 + 9 + 7) - (8 + 2 + 0)$$

$$(21) - (10) = 11$$

$\therefore 589270$ is divisible by 2, 5, 10 and 11

$$\frac{589270}{11}$$

57. (b) $\frac{249K876}{2 \times 3}$

Sum of digits should be divisible by 3 since it is already divisible by 2

$$\frac{36 + K}{3}$$

$\therefore$ Least value of $K = 0$

58. (c) $\frac{217924K}{6}$

$$6 \Rightarrow 2 \times 3$$

$\therefore$ Sum of digits should be divisible by 3 and K will be a even no

$$\frac{25 + K}{3}$$

$\therefore K = 2$

59. (a) Sum of digits should be divisible by 9

$$\frac{57 \times 716}{9}$$

$$\Rightarrow \frac{26 + x}{9}$$

Least value of $x = 1$